



ASX Announcement
27 September 2019

Halls Creek Project Mineral Resource & Ore Reserve Update

Pantoro Limited (**ASX:PNR**) (**Pantoro**) is pleased to provide its annual Mineral Resource and Ore Reserve statement for the Halls Creek Project as at 31 May 2019. The project has seen year on year upgrade to the Mineral Resource including the addition of Mineral Resources at Grants Creek for the first time. The Ore Reserve has shown depletion aligned to mining at the Nicolsons gold mine and addition to the Ore Reserve at Wagtail Mine.

Mineral Resources related to the Norseman Gold Project have been reported separately in a release to the ASX on 9 September 2019 in an announcement titled 'Successful Start to Norseman Drilling Campaign'. The Norseman transaction closed after the reporting date.

Key Highlights

- The total Halls Creek Project Mineral Resource now stands at 1,631 million tonnes @ 7.5 g/t for 394,000 ounces, maintaining the inventory in line with the previous years result after mining depletion and pillar sterilisation.
- The total Ore Reserve now stands at 844,000 tonnes @ 6.59 g/t for 179,000 ounces, an 18 % decrease after mining depletion and pillar sterilisation.
- The Nicolsons Ore Reserve has been extended to 420 metres below surface, with the Inferred Mineral Resource extending to approximately 500 metres below surface. Drilling has continued to return high grade results down plunge to 500 metres depth, indicating strong potential for ongoing Ore Reserve updates as development and drilling at the mine progress deeper.
- Maiden Inferred Mineral Resource for Perseverance and Star of Kimberley deposits at Grants Creek of 179,000 tonnes @ 2.4 g/t for 14,000 ounces. Additional drilling to continue growth of Mineral Resources is planned for Grants Creek.
- Decline development at Wagtail had progressed to access the third sub level by the report date, with limited minor ore development in the period. Ore development since the reporting date has had a positive reconciliation against Ore Reserve providing confidence for ongoing upgrades as the Wagtail mine development and drilling progresses.
- In mine, near mine and regional exploration and resource development drilling is ongoing at Nicolsons, Grants Creek and Mary River. This work will continue to focus on growth of the Mineral Resource and Ore Reserve inventory to be mined during the coming years.

Key Mineral Resource details are set out in the table below:

	Measured			Indicated			Inferred			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
Nicolsons Underground	146	15.9	75	239	9.4	72	91	10.4	31	476	11.6	177
Nicolsons Open Pits	64	10.5	22	146	5.0	23	46	4.5	7	256	6.3	51
Wagtail	-	-	-	494	7.4	118	126	6.7	27	620	7.3	145
Grants Creek	-	-	-	-	-	-	179	2.4	14	179	2.4	14
LG Stockpiles	100	1.9	6	-	-	-	-	-	-	100	1.9	6
Total	310	10.3	102	879	7.5	213	442	5.5	78	1631	7.5	394

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Key Ore Reserve details are set out in the table below:

	Proven			Probable			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
Nicolsons Underground	149	9.0	43	162	7.7	40	311	8.3	83
Nicolsons Open Pits	31	12.4	12	23	8.4	6	54	10.7	19
Wagtail	-	-	-	379	5.9	72	379	5.9	72
LG Stockpiles	100	1.9	6	-	-	-	100	1.9	6
Total	280	6.8	61	564	6.5	118	844	6.6	179

Notes: Nicolsons Underground (2.5 g/t cut off grade applied).

Nicolsons Open Pits (0.6 g/t cut off grade applied).

Wagtail includes the Rowdies deposit. (2.5 /gt cut off grade applied).

Grants Creek includes Perserverance and Star of Kimberly (1.0 g/t cut off grade applied).

Commenting on the Mineral Resource and Ore Reserve upgrade, Managing Director Paul Cmrlec said:

“The Nicolsons Project has provided excellent outcomes for Pantoro during recent years. The year on year replacement of Mineral Resources provides confidence for ongoing operations at the mine into the future. With extensive capital re-investment in the development of the Wagtail Underground mine during the year, the mine is set to continue to deliver strong cashflows during the coming year. We have seen positive ore reconciliation from Wagtail to date and will strive to achieve growth in Ore inventories at Wagtail, in line with the growth delivered at Nicolsons in recent years.

With the ongoing exploration effort both within the existing mines and in the wider project areas, we are confident that Mineral Resource inventories will continue to grow into the future.”

Enquiries

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Mineral Resource and Ore Reserve Update

The Mineral Resource was completed in accordance with JORC 2012 by independent resource consulting group Entech with data provided by Pantoro Geologists and under the supervision and review of the Competent Person. The Ore Reserve was completed by Pantoro Mining Engineers under the direction of the Competent Person. Additional Ore has been identified using drilling data, combined with mine development activities. The Mineral Resource and Ore Reserve inventory has been adjusted for depletion and pillar exclusion as a result of mining. The new Mineral Resource and Ore Reserve is compared with the previous estimate in the charts and table below. The previous Mineral Resource and Ore Reserve estimate was released to the ASX on the 2 August 2018 in a release titled 'Nicolsons Project Mineral Resource & Ore Reserve Update'.

The new Mineral Resource and Ore Reserve update is calculated as at the 31 May 2019. A comparison with the previous Mineral Resource Estimate and Ore Reserve Calculation including mining depletion is provided below.



Mineral Resource Update – Nicolsons

The Nicolsons Mineral Resource update was completed in accordance with JORC 2012 by independent resource consulting group Entech with data provided by Pantoro Geologists and under the supervision and review of the Competent Person. Key changes in the Mineral Resource Estimate include:

- Modification of ore zones – the geological interpretation of the mineralised domains for the Johnston and Anderson lodes were re-interpreted using additional face sampling and drilling data obtained since the previous Mineral Resource and Ore Reserve upgrade.
- Additional data – 27,042 metres drilling from 256 reverse circulation and diamond holes were completed during the year, with all available assay results utilised in the Mineral Resource Estimate. Face sampling of ore drives and rises was also undertaken with 2,047 faces sampled for 6,674 individual sample utilised in the estimate.
- Re-classification of ore categories – ore within the block model was classified into the Measured, Indicated and Inferred categories as appropriate. The classification was based on data density and statistical analysis of that data.
- Depletion of zones that have been mined or sterilised.

Details of the Nicolsons Mineral Resource are set out in the table below:

	Measured			Indicated			Inferred			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
May 2018 Resource	222	15.6	111,0	432	7.5	104,0	163	9.0	47,0	817	10.0	262.0
May 2019 Resource	210	9.5	96	385	7.7	95.0	137	8.4	37	732	9.68	228.0

Ore Reserve Update – Nicolsons

The Nicolsons Ore Reserve has been completed utilising functional mine designs using both underground and open pit methods. The Open Pit Ore Reserve (calculated at A\$1,400/oz) has not been revised since the previous update, and remains unchanged. The Open Pit Ore Reserve and accompanying JORC Table 1 is detailed in a release to the ASX on the 1st June 2017 in a release titled “Mineral Resource and Ore Reserve Upgrade Demonstrates Strong Growth Potential at Nicolsons”.

Changes to the Nicolsons Ore Reserve include:

- Addition of Ore Reserve blocks where the Mineral Resource estimate has classified new zones of Measured and Indicated Mineral Resource material as a result of additional data.
- Modification of the tonnage and grade of ore blocks where the revised block model has changed since the last estimate.
- Depletion of zones that have been mined or left in pillars.
- Adjustment of ore stockpiled on the surface.

Details of the Nicolsons Ore Reserve are set out in the table below:

	Proven			Probable			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
May 2018 Nicolsons Underground	198	11.8	75	275	6.5	58	473	8.8	133
May 2018 Nicolsons Pits	31	12.4	12	23	8.4	6	54	10.7	19
May 2019 Nicolsons Underground	149	9.0	43	162	7.7	40	311	8.3	83
May 2019 Nicolsons Pits	31	12.4	12	23	8.4	6	54	10.7	19

Mineral Resource Update – Wagtail

The Wagtail Mineral Resource update has been completed in accordance with JORC 2012 by independent resource consultant Entech under the supervision and review of the Competent Person. The Wagtail Mineral Resource is detailed in a release to the ASX on 5 March 2019 with only the Rowdies ore zones being updated and re estimated to reflect the additional focused drilling undertaken on the down plunge extensions of the deposit. Ore zones from the Wagtail South and Wagtail North are unchanged from the May 2018 Mineral Resource estimate, and has not changed since the time of that report (with the exception of mining development depletion up to May 30 2019). Key changes detailed in this report include:

- Inclusion of additional Mineral Resource in the Rowdies ore zone through additional drilling programs.
- Depletion of Ore mined via underground development up to May 30 from the model.

	Measured			Indicated			Inferred			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz	Tonnes	Grade	Ounces
May 2018 Wagtail	-	-	-	450	6.8	98	124	7.0	28,	574	6.8	126
May 2019 Wagtail	-	-	-	494	7.4	118	126	6.7	27	620	7.3	145

Ore Reserve Update – Wagtail

The Wagtail Ore Reserve is based on functional underground mine designs. The Wagtail Ore Reserve is detailed in a release to the ASX on 2 April 2019, and has not changed since the time of that report. Key changes detailed in that report include:

- Inclusion of additional Ore Reserve blocks in the Rowdies ore zone where the Mineral Resource estimate has classified new zones of Measured and Indicated Mineral Resource material as a result of additional data.
- Depletion of Ore mined via underground development up to May 31 2019 from the model.

	Proven			Probable			Total		
	kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
May 2018 Wagtail	-	-	-	356	5.4	62	356	5.4	62
May 2019 Wagtail	-	-	-	379	5.9	72	379	5.9	72

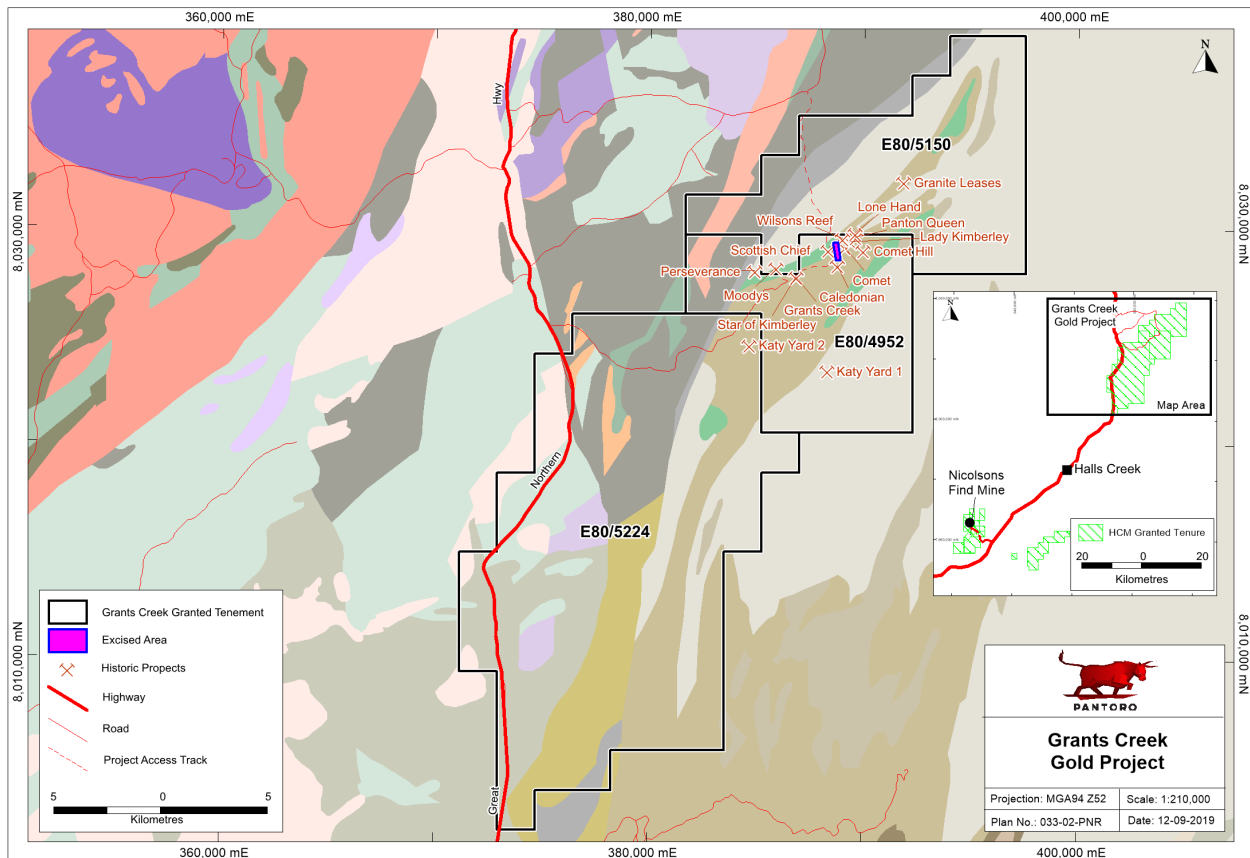
Maiden Mineral Resource – Grants Creek

The Grants Creek maiden Mineral Resource update was completed in accordance with JORC 2012 by independent resource consulting group Entech with data provided by Pantoro geologists and under the supervision and review of the Competent Person. The Grants Creek Mineral Resource encompasses the Perseverance and Star of Kimberley deposits which were drilled over 2 campaigns in 2018 and 2019 field seasons following up on historic workings and drilling.

The current resource forms the basis for further work at the current prospects but also the broader project with the consolidation of the project areas addition of two new tenements E80/5150 and E80/5224 encompassing all the historic workings in the area as well as several prospective greenfield targets identified by review of the available geophysics.

The plan for future work is to complete reconnaissance over the broader project area to refine further drill targets which will be tested in conjunction with further work at the Star of Kimberley and Perseverance deposits to infill and extend the already identified mineralisation. The Material information required as ASX listing rule 5.8.1 is contained in the appendices to this announcement.

	Cut off Au g/t	Indicated			Inferred			Total		
		kT	Grade	kOz	kT	Grade	kOz	kT	Grade	kOz
Perseverance	1.0	-	-	-	154	2.3	11.5	154	2.3	11.5
Star of Kimberley	1.0	-	-	-	25	3.0	2.5	25	3.0	0.2
Total		-	-	-	179	2.4	14	179	2.4	14



Grants Creek Tenement Plan

Material Summary - Nicolsons Gold Mineral Resource Update

Material information summary as required under ASX Listing Rule 5.8 and JORC 2012 reporting guidelines.

Mineral Resource Statement

The Mineral Resource Statement for the Nicolsons Deposit Gold Mineral Resource Estimate (MRE) was prepared during July 2019 and is reported according to the Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves (the 'JORC Code') 2012 edition.

This MRE update includes an additional 27,042 m drilling from 256 reverse circulation and diamond holes as well as 6,674 m of sampling from 2,047 underground production faces, since the 2018 MRE (July 2018) and is reported excluding all historical and recent mining activity, surveyed up to the 31st of May 2019. Depth from surface to the current vertical limit of the Mineral Resource is approximately 500 m.

In the opinion of Pantoro, the resource evaluation reported herein is a reasonable representation of the global gold mineral resources within the deposits, based on Reverse Circulation Drilling, Diamond Drilling and Underground Production sampling data available as of June 13th, 2019. The MRE comprises oxide, transitional and fresh material and is detailed on page 1 of this announcement.

Interpretations for Nicolsons were informed by Reverse Circulation drilling (423 drill holes), with Diamond Drilling (701 drill holes inclusive of diamond tails) and Face Sampling (5,250 faces), for 3,278 m of drilling intersecting ore in updated domains and a combined total of 134,797 m of drilling from 1,124 drill holes and 18,835 m of sampling from 5,250 production faces.

This MRE comprises Inferred Mineral Resources which are unable to have economic considerations applied to them, nor is there certainty that they will be converted to Measured or Indicated Resources through further sampling.

Drilling Techniques

A variety of drilling techniques were used to test the Nicolson's deposit, however the recent drilling has utilised diamond drilling, HQ3, NQ2 and LTK48 diameter core from Reverse Circulation pre-collars or underground drill caddies. All pre-collars were sampled. Reverse circulation drilling was carried out using a face sampling hammer and a 140-146 mm with historical drilling being 130 mm diameter bit.

Sampling Techniques, Sub-Sampling Techniques and Sample Preparation

Diamond

All diamond core is orientated and logged by a qualified geologist. It is sampled according to geology, with only selected samples assayed. Core is cut in half under the supervision of an experienced geologist utilising an Almonte diamond core-saw, with the RHS of cutting line routinely assayed, the other half retained in core trays on site for further analysis and storage. All mineralised zones are sampled as well as material considered barren either side of the mineralised interval. Samples are a maximum of 1.2m, with shorter intervals utilised according to geology to a minimum interval of 0.15m where clearly defined mineralisation is evident. All diamond core is stored in core trays and is aligned, measured and marked up in metre intervals referenced back to downhole core blocks recording run meterage and any core loss if encountered. Downhole surveys are conducted during drilling using a reflex electronic single shot cameras at collar, 20 m then every 30 m thereafter. No significant core loss has been noted from recent drilling. Visible gold is encountered at the project and where observed during logging, Screen Fire Assays are conducted

Reverse Circulation

Samples are collected via both a cone splitter and a rig-mounted static splitter used, with sample falling through a riffle splitter and sampled every 1m. Current Nicolson's diamond hole pre-collars are sampled on 2m composites with 1m splits retained for further assays as required.

All RC holes are geologically logged by a qualified geologist and logging parameters include: depth from, depth to, condition, weathering, oxidation, lithology, texture, colour, alteration style, alteration intensity, alteration mineralogy, sulphide content and composition, quartz content, veining, and general comments. 100% of the holes are logged. Appropriately qualified company personnel supervise the drilling programs on site and monitor sample quality and integrity. Recovery and sample quality were visually monitored and laboratory sample weights recorded and reviewed. Chip trays from each logged interval are retained and stored for reference. No significant water was encountered and are samples are typically dry.

Reverse Circulation samples of 2-5kg in weight are dispatched to an external accredited laboratory Bureau Veritas in Perth (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). Diamond samples 0.5-3.5kg samples are dispatched to an external accredited laboratory (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). The processes applied are industry standard for this type of sample.

Historical holes - RC drilling was used to obtain 1 m samples from which 2 - 3 kg was crushed and sub-split prior to pulverisation and then a 40 g aliquot for fire assay. Review of drilling programmes indicate all intervals were assayed and is considered to be to industry standard at that time

Face Sampling

Face Chips samples are nominally chipped perpendicular to mineralisation across the face from left to right, and sub-set via geological features as appropriate.

Sample Analysis Method

Samples were analysed at Bureau Veritas in Perth. Gold assays are determined using fire assay with 40g charge. Where other elements are assayed using either AAS base metal suite or acid digest with ICP-MS finish. Screen fire assays consists of screening 500g of the sample to 106 microns. The plus fraction is fire assayed for gold and a duplicate assay is performed on the minus fraction. The size fraction weights, coarse and fine fraction gold content and total gold content are reported. The methods used approach total mineral consumption and are typical of industry standard practice.

CRM standards, blanks and repeats are included as part of the QAQC system. In addition the laboratory has its own internal QAQC comprising standards, blanks and duplicates. Sample preparation checks of pulverising at the laboratory include tests to check that the standards of 90% passing 75 micron is being achieved. Follow-up re-assaying is performed by the laboratory upon company request following review of assay data. Acceptable bias and precision is noted in results given the nature of the deposit and the level of classification.

No assay data was adjusted.

Geology and Geological Interpretation

The Nicolsons deposit is located within the Halls Creek Orogen (HCO) in the Kimberley region of Western Australia. Deposits are associated with north-northeast trending shear zones which form part of the Central belt of the HCO.

Gold mineralization in the Nicolson's Find area is structurally controlled within the 400 m wide NNE trending dextral strike slip Nicolson's Find Shear Zone (NFSZ) and is hosted within folded and metamorphosed turbiditic greywackes, felsic volcanoclastics, mafic volcanics and laminated siltstones and mudstones. This zone forms part of a regional NE trending strike slip fault system developed across the Halls Creek Orogen. Mineralisation is strongly correlated with quartz veining and with Fe-Si-K alteration halos developed in the wall rocks to the veins.

Estimation Methodology

Two domains were updated during the 2019 Nicolsons MRE, these being Domain 4 (Johnston Lode) and Domain 49 (Anderson Lode), all other domains remain as per the 2018 Nicolsons MRE.

A two dimensional (2D) Ordinary Kriging interpolation approach was employed to estimate block grades. The 2D interpolation approach utilised varies from a three-dimensional approach (3D) in that estimation of both an accumulation variable (intercept gold composite weighted by true width) and the true width variable, is undertaken in a 2D plane.

The gold mineralisation is hosted within multiple lodes within narrow quartz veins in the corridor of the main Nicolsons shear zone (NSZ) and the interpreted mineralised domains were utilized as hard boundaries within the estimation process. Top cuts were applied to the gram-meter accumulation variable after statistical, spatial analysis and assessment of percentage of metal reduction within each mineralized domain with cut values being:

- Domain 4 = 160 g/m Accumulation and a 6.5% metal reduction,
- Domain 49 = 160 g/m Accumulation and a 11.1% metal reduction.

The 2D parent estimation block size selected for interpolation was 10 metres in the Y and X direction with the parent block size being determined through kriging neighborhood analysis, review of vein dimensions, drilling density and mining selectivity. Block sub-celling size was selected for appropriate volume fill within the mineralization wireframes. No block rotation was applied.

For the updated domains (4 and 49) variography was conducted in the plane of mineralisation and from which parameters for the Ordinary Kriging were derived and applied to each individual domain. The search strategy was a maximum extrapolation distance of 120 m and 135 m over three search passes for Domains 4 and 49 respectively. The first pass search was equal to the variogram maximum range (40 m for Domain 4 and 45 m for Domain 49) with the second pass search double the variogram range (80 m for Domain 4 and 90 m for Domain 49) and the third pass triple the variogram range (120 m for Domain 4 and 135 m for Domain 49). A constant minimum of 4 and maximum of 12 composites was maintained across all search passes for Domain 4. Domain 49 utilised a minimum of 4 and a maximum of 12 for the first pass, dropping to a minimum of 3 samples for the second and third passes.

A distance based high-grade limit function was applied to Domain 49, limiting accumulation values above 60 gram-metres to half the variogram range (22.5 m).

Check estimates were completed utilising Inverse Distance weighting.

Global and local validation of the gold variable estimated outcomes was undertaken with statistical analysis, swath plots and visual comparison (cross and long section) against input data.

Classification Criteria

This current Mineral Resource Estimate has been classified as Measured, Indicated and Inferred to appropriately represent confidence and risk with respect to data quality, drill hole spacing, geological and grade continuity, mineralisation volumes, recent and historical mining activity as well as metal distribution. Additional considerations were the stage of project assessment, amount of diamond drilling, current understanding of mineralisation controls and selectivity within an underground mining environment. The Nicolson deposit has been mined continuously by Underground methods since mid-2015 with recent data from underground production supporting both grade and geological continuity. The bulk of the data utilised in the current Mineral Resource estimate is from recently acquired drilling and sampling with an additional 27,042 m drilling from 256 reverse circulation and diamond holes as well as 6,674 m of sampling from 2,047 underground production faces.

Blocks in the resource model have been allocated a confidence category of Inferred or Indicated based on a combination of various estimation quality derived parameters, data support, data quality, mineralization continuity and mining knowledge.

The reported Mineral Resource was constrained at depth by the available drill hole spacing outlined for Inferred classification, nominally 500 m below surface, with the threshold for Open Pit and Underground reporting being nominally 100 m below surface.

Cut-Off Parameters

The global gold Mineral Resource has been reported at a 2.5 g/t gold cut-off for Underground (greater than 100 m below topographic surface) resources and a 0.6 g/t cutoff for Open Pit (within 100 m of topographic surface) resources and is based upon economic parameters currently utilised at the Nicolson operations, where deposits of the same style, commodity, comparable size and mining methodology are currently being extracted. Tonnages were estimated on a dry basis.

Assessment of Reasonable Prospects for Eventual Economic Extraction

The material reported in the Nicolson MRE is considered to meet Reasonable Prospects for Eventual Economic Extraction based on the following considerations.

Similar mineralised material is currently mined, at the reported cut-off, in the most recent underground levels at Nicolson. This includes fresh material mined and processed in the current Nicolson processing facility where recoveries have been consistently achieving 92% and support recovery of the in situ Mineral Resource via conventional gravity and cyanidation methodology.

The MRE extends nominally 500 m below topographic surface and lies within 100 vertical metres of active level development. Pantoro considers material at this depth would fall within the definition of 'reasonable prospect of eventual economic extraction' within an underground mining framework.

No dilution, cost factors or metallurgical recovery factors were applied to the Mineral Resources or Resource Tabulations.

Material Summary – Grants Creek Maiden Mineral Resource Update

Material information summary as required under ASX Listing Rule 5.8 and JORC 2012 reporting guidelines.

Mineral Resource Statement

The Maiden Mineral Resource Statement for the Grants Creek Mineral Resource Estimate (MRE) was prepared during July 2019 and is reported according to the Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves (the 'JORC Code') 2012 edition.

This Maiden Resource is reported excluding known historical mining activity, as of 31st of May 2019. Depth from surface to the current vertical limit of the Mineral Resource is approximately 100 m.

In the opinion of Pantoro, the resource evaluation reported herein is a reasonable representation of the global gold mineral resources within the deposits, based on Reverse Circulation Drilling, Diamond Drilling data available as of June 24th, 2019. The MRE is detailed on page 5 of this announcement.

Interpretations for Grants Creek were informed by Reverse Circulation drilling (159 drill holes), with Diamond Drilling (5 drill holes inclusive of diamond tails) for 352 m of drilling intersecting ore and a combined total of 7728 m of drilling from 164 drill holes.

This MRE comprises Inferred Mineral Resources which are unable to have economic considerations applied to them, nor is there certainty that they will be converted to Measured or Indicated Resources through further sampling.

Drilling Techniques

A variety of drilling techniques were used to test the Perserverance and Star of Kimberley deposits, however the recent drilling has utilised predominantly Reverse Circulation and 3 NQ2 diamond drillholes from Reverse Circulation pre-collars. All pre-collars were sampled. Reverse circulation drilling was carried out using a face sampling hammer and a 140-146 mm with historical drilling being 130 mm diameter bit.

Sampling Techniques, Sub-Sampling Techniques and Sample Preparation

Diamond

All diamond core is orientated and logged by a qualified geologist. It is sampled according to geology, with only selected samples assayed. Core is cut in half under the supervision of an experienced geologist utilising an Almonte diamond core-saw, with the RHS of cutting line routinely assayed, the other half retained in core trays on site for further analysis and storage. All mineralised zones are sampled as well as material considered barren either side of the mineralised interval. Samples are a maximum of 1.2m, with shorter intervals utilised according to geology to a minimum interval of 0.15m where clearly defined mineralisation is evident. All diamond core is stored in core trays and is aligned, measured and marked up in metre intervals referenced back to downhole core blocks recording run meterage and any core loss if encountered. Downhole surveys are conducted during drilling using a reflex electronic single shot cameras at collar, 20 m then every 30 m thereafter. No significant core loss has been noted from recent drilling.

Reverse Circulation

Samples are collected via a rig-mounted cone splitter used, and sampled every 1m. Diamond hole pre-collars are sampled every metre downhole.

All Pantoro RC holes are geologically logged by a qualified geologist and logging parameters include: depth from, depth to, condition, weathering, oxidation, lithology, texture, colour, alteration style, alteration intensity, alteration mineralogy, sulphide content and composition, quartz content, veining, and general comments. 100% of the holes are logged. Appropriately qualified company personnel supervise the drilling programs on site and monitor sample quality and integrity. Recovery and sample quality were visually monitored and laboratory sample weights recorded and reviewed. Chip trays from each logged interval are retained and stored for reference. No significant water was encountered and are samples are typically dry. Lithological data from historical drilling includes metadata recovered from previous drilling.

Reverse Circulation samples of 2-5kg in weight are dispatched to an external accredited laboratory Bureau Veritas in Perth (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). Diamond samples 0.5-3.5kg samples are dispatched to an external accredited laboratory (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). The processes applied are industry standard for this type of sample.

Historical holes - RC drilling was used to obtain 1 m samples from which 2 - 3 kg was crushed and sub-split prior to pulverisation and then a 40 g aliquot for fire assay. Review of drilling programmes indicate all intervals were assayed and is considered to be to industry standard at that time

Sample Analysis Method

Samples were analysed at Bureau Veritas in Perth. Gold assays are determined using fire assay with 40g charge. Where other elements are assayed using either AAS base metal suite or acid digest with ICP-MS finish. Screen fire assays consists of screening 500g of the sample to 106 microns. The plus fraction is fire assayed for gold and a duplicate assay is performed on the minus fraction. The size fraction weights, coarse and fine fraction gold content and total gold content are reported. The methods used approach total mineral consumption and are typical of industry standard practice.

CRM standards, blanks and repeats are included as part of the QAQC system. In addition the laboratory has its own internal QAQC comprising standards, blanks and duplicates. Sample preparation checks of pulverising at the laboratory include tests to check that the standards of 90% passing 75 micron is being achieved. Follow-up re-assaying is performed by the laboratory upon company request following review of assay data. Acceptable bias and precision is noted in results given the nature of the deposit and the level of classification.

No assay data was adjusted.

Geology and Geological Interpretation

The Grants Creek gold mineralisation is hosted by quartz reefs within deformed and folded metasedimentary and metavolcanic rocks of Proterozoic age. The oldest rocks of the complex are the Ding Dong Downs Volcanics and the Sophie Downs Granite separated from the overlying Halls Creek Group by an unconformity.

The project area also covers part of the Lower Proterozoic Halls Creek Group sediments and sub-volcanics of the Lamboo Complex whilst the Biscay and overlying Olympio Formations comprise the Upper Halls Creek Group. Overlying this Group, the White Water Volcanics Formation is also present to the east of the Halls Creek Fault Zone, a major structural feature that trends northeast across the Grants Creek leases.

Estimation Methodology

Gold mineralisation is hosted within multiple lodes within quartz veins in proximity to the Halls Creek Fault zone and the interpreted mineralised domains were utilized as hard boundaries within the estimation process. Utilising geology and grade to underpin mineralisation interpretation three domains (10,11 and 12) were defined in the Perseverance deposit and one domain in the Star of Kimberley deposit (40).

A three dimensional (3D) Ordinary Kriging interpolation approach was employed to estimate block grades within the mineralisation domains, underpinned by composites on 2 metre lengths. Composites included all available diamond, reverse circulation assay data and were 'best fit' with residuals reviewed and discarded prior to estimation.

Top cuts were applied to the composited gold variable after statistical, spatial analysis and assessment of percentage of metal reduction within each mineralized domain were completed. Based on the analysis a top cut of 14 g/t Au was applied to Domain 10 (Perseverance). No other domains were top cut.

The 3D parent estimation block size selected for interpolation was 10 metres in the Y, X and Z direction with the parent block size being determined through kriging neighbourhood analysis, review of vein dimensions, drilling density and potential mining selectivity. Block sub-celling size was selected for appropriate volume fill within the mineralization wireframes. No block rotation was applied.

Variography was conducted in the plane of mineralisation and from which parameters for the Ordinary Kriging and search neighbourhoods were derived and applied to each individual domain. The optimised search strategy was single pass with a maximum extrapolation distance of 65 m and 45 m for Perseverance and Star of Kimberley respectively. The search ranges were nominally equivalent to the variogram maximum range with a minimum of 4 and maximum of 10 composites maintained across all Domains.

Check estimates were completed utilising Inverse Distance weighting. Global and local validation of the gold variable estimated outcomes was undertaken with statistical analysis, swath plots and visual comparison (cross and long section) against input data.

Classification Criteria

The Grants Creek MRE has been classified as Inferred to appropriately represent confidence and risk with respect to data quality, drill hole spacing, geological and grade continuity, mineralisation volumes, historical mining activity as well as metal distribution. Additional considerations were the stage of project assessment, amount of diamond drilling, current understanding of mineralisation controls and selectivity within an open pit mining environment.

Blocks in the resource model have been allocated a confidence category of Inferred based upon a combination of various estimation quality derived parameters, data support (nominally less than 30m), data quality and mineralization continuity.

The reported Mineral Resource was constrained at depth by the available drill hole spacing outlined for Inferred classification (30m) and the assessment of Reasonable Prospects for Eventual Economic Extraction (RPEEE), 100m below topographic surface.

Cut-Off Parameters

The global gold Mineral Resource has been reported at a 1.0 g/t cutoff for Open Pit and is based upon economic parameters and depths (within 100 m of topographic surface) currently utilised at the nearby Nicolsons operations, where deposits of the same style, commodity, comparable size and mining methodology are currently being extracted.

Tonnages were estimated on a dry basis.

Assessment of Reasonable Prospects for Eventual Economic Extraction

The material reported in the Grants Creek Maiden MRE is considered to meet Reasonable Prospects for Eventual Economic Extraction based on the following considerations.

Metallurgical test work on fresh and transitional samples returned an average of 92% and support recovery of the in situ Mineral Resource via conventional gravity and cyanidation methodology.

The MRE extends nominally 100 m below topographic surface. Pantoro considers material at this depth would fall within the definition of 'reasonable prospect of eventual economic extraction' within an open pit mining framework, based upon comparisons with the nearby Nicolsons operations where deposits of the same style, commodity, comparable size and mining methodology are currently being extracted.

No dilution, cost factors or metallurgical recovery factors were applied to the Mineral Resources or Resource Tabulations.

Compliance Statements

Exploration Targets, Exploration Results, Mineral Resources

The information in this report that relates to Exploration Targets, Exploration Results and Mineral Resources is based on information compiled by Mr Scott Huffadine (B.Sc. (Hons)), a Competent Person who is a Member of the Australian Institute of Geoscientists. Mr Huffadine is a Director and full time employee of the company. Mr Huffadine is eligible to participate in short and long term incentive plans of and holds shares, options and performance rights in the Company as has been previously disclosed. Mr Huffadine has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Huffadine consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

Ore Reserves

The information in this report that relates to Ore Reserves is based on information compiled by Mr Paul Cmrlec (B. Eng (Mining) (Hons), a Competent Person who is a Member of the Australian Institute of Mining and Metallurgy. Mr Cmrlec is a Director and full time employee of the company. Mr Cmrlec is eligible to participate in short and long term incentive plans of and holds shares, options and performance rights in the Company as has been previously disclosed. Mr Cmrlec has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Cmrlec consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

APPENDIX A – JORC CODE 2012 EDITION – TABLE 1

SECTION 1: SAMPLING TECHNIQUES AND DATA – NICOLSONS

Criteria	JORC Code explanation	Commentary
Sampling techniques	- Nature and quality of sampling (eg cut channels, random chips, or specific specialised industry standard measurement tools appropriate to the minerals under investigation, such as down hole gamma sondes, or handheld XRF instruments, etc). These examples should not be taken as limiting the broad meaning of sampling. - Include reference to measures taken to ensure sample representivity and the appropriate calibration of any measurement tools or systems used. - Aspects of the determination of mineralisation that are Material to the Public Report. - In cases where 'industry standard' work has been done this would be relatively simple (eg 'reverse circulation drilling was used to obtain 1 m samples from which 3 kg was pulverised to produce a 30 g charge for fire assay'). In other cases more explanation may be required, such as where there is coarse gold that has inherent sampling problems. Unusual commodities or mineralisation types (eg submarine nodules) may warrant disclosure of detailed information.	- This report relates to the annual update of the Mineral Resource and Ore Reserve statement for Nicolson's Find (Nicolson's) deposit at the Nicolson's gold project. - The Nicolson's deposit has been sampled by RC, underground diamond, historical earth saw lines within the existing open pit, underground face sampling, minor Kempe Diamond drilling and minor historical RAB about the Nicolson's open pit area. - For RC drilling, measures taken to ensure sample representivity include the presence of a geologist at the rig whilst drilling, cleaning of the splitter at the end of every 3 m drill string, confirmation that drill depths match the accompanying sample interval with the drilling crew and the use of duplicate and lab/blank standards in the drilling programme. - Face Sampling,, each development face / round is chip sampled perpendicular to mineralisation. The sampling intervals are dominated by geological constraints (e.g. rock type, veining and alteration / sulphidation etc.). The majority of exposures within the orebody are sampled - For surface diamond drilling, measures taken include regular survey of drill holes, cutting of core along the orientation line where possible, and half core is submitted to an accredited laboratory. Industry standard blanks and standards are also submitted and reported by the laboratory. Drilling is completed in HQ3 or NQ2. HQ3 or NQ2 core is logged and sampled according to geology, with only selected samples assayed. Core is halved, with one side assayed, and the other half retained in core trays on site for further analysis. Samples are a maximum of 1m, with shorter intervals utilised according to geology. - For underground diamond drilling, measures taken include regular survey of drill holes, cutting of core along the orientation line where possible, and half core is submitted to an accredited laboratory. Industry standard blanks and standards are also submitted and reported by the laboratory. Drilling is completed in LTK 60. LTK 60 core is logged and sampled according to geology, with only selected samples assayed. Core is halved, with one side assayed, and the other half retained in core trays on site for further analysis. Samples are a maximum of 1m, with shorter intervals utilised according to geology. - Kempe Diamond drill core (LTK48 diameter) was hole core sampled ie all of the core was sampled and assayed.

Criteria	JORC Code explanation	Commentary
		- No information has been recorded for historic sampling of the earth saw trenches and RAB in terms of the sample sizes and method of splitting. The lack of the information is not considered material to the estimation. - Historical holes - RC and aircore drilling was used to obtain 1 m samples from which 2 - 3 kg was crushed and sub-split to yield 250 for pulverisation and then a 40 g aliquot for fire assay. Upper portions of deeper holes were composited to 3m sample intervals and sub-split to 1 m intervals for further assay if an anomalous composite assay result was returned. For later drilling programs all intervals were assayed.
Drilling techniques	Drill type (eg core, reverse circulation, open-hole hammer, rotary air blast, auger, Bangka, sonic, etc) and details (eg core diameter, triple or standard tube, depth of diamond tails, face-sampling bit or other type, whether core is oriented and if so, by what method, etc).	- RC drilling was completed with several rigs. All RC rigs used face sampling hammers with bit size of 140 – 146mm. Historical holes used a 130 mm bit size). Aircore drilling was completed by the RC rig with an aircore bit assembly. - HQ 3 Diamond drilling was conducted in 2014 for geotechnical and assay data. Diamond holes were oriented using a Reflex orientation tool. Diamond holes were geologically and geotechnical logged. - Underground face samples, were chipped from the desired domain(rock type) using an Estwing geology hammer. A number of chips were taken between knee and head height from the geological domain to obtain a representative sample. The chips were put in a pre numbered sample bags. - Earth saw trenches were used to grade control the historic Nicolson's pit the trenches were sampled at meter intervals. No other information was recorded for the method. - LTK60 core is drilled with both an Atlas Copco carrier mounted U8 and fixed U6 DH Rig With Rod Handler and wire line. - LTK48 is drilled using a Kempe U2 Rig air rig
Drill sample recovery	- Method of recording and assessing core and chip sample recoveries and results assessed. - Measures taken to maximise sample recovery and ensure representative nature of the samples. - Whether a relationship exists between sample recovery and grade and whether sample bias may have occurred due to preferential loss/gain of fine/coarse material.	- All holes were logged at site by an experienced geologist. Recovery and sample quality were visually observed and recorded. Recovery for older (pre 2011) holes is unknown. - All drilling was completed within rig capabilities. Surface Rigs used auxiliary air boosters when appropriate to maintain sample quality and representivity. Where aircore drilling could not provide sufficient penetration an RC drilling set-up was used. - There is no known relationship between recovery and grade. Diamond drilling of oxide and transitional material in previous campaigns noted high core loss in mineralised zones. No core loss was noted in fresh material. Good core recovery has generally been achieved in all sample types in the current drilling programs.

Criteria	JORC Code explanation	Commentary
Logging	- Whether core and chip samples have been geologically and geotechnically logged to a level of detail to support appropriate Mineral Resource estimation, mining studies and metallurgical studies. - Whether logging is qualitative or quantitative in nature. Core (or costean, channel, etc) photography. - The total length and percentage of the relevant intersections logged.	- Geological logging parameters include: depth from, depth to, condition, weathering, oxidation, lithology, texture, colour, alteration style, alteration intensity, alteration mineralogy, sulphide content and composition, quartz content, veining, and general comments. - Underground development faces are mapped geologically. - Geotechnical logging of diamond holes included the recording of recovery, RQD, structure type, dip, dip direction, alpha and beta angles, shape, roughness and fill material of fractures - All drill chips were logged on 1 m increments, the minimum sample size. A subset of all chip samples is kept on site for reference. - Diamond drilling was logged to geological boundaries and is considered quantitative. Core was photographed. - All drilling has been logged apart from diamond drill pre-collars.
Sub-sampling techniques and sample preparation	- If core, whether cut or sawn and whether quarter, half or all core taken. - If non-core, whether riffled, tube sampled, rotary split, etc and whether sampled wet or dry. - For all sample types, the nature, quality and appropriateness of the sample preparation technique. - Quality control procedures adopted for all sub-sampling stages to maximise representivity of samples. - Measures taken to ensure that the sampling is representative of the in situ material collected, including for instance results for field duplicate/second-half sampling. - Whether sample sizes are appropriate to the grain size of the material being sampled.	- Core samples were sawn in half with one half used for assaying and the other half retained in core trays on site for future analysis. - RC drill chip samples were collected with either a three-tier, rotary or stationary cone splitter depending on the drill rig used. Aircore drill samples were subset using a 3 tier riffle splitter. Most (> 95%) of samples are recorded as being dry. - Face Chips samples are nominally chipped perpendicular to mineralisation across the face from left to right, and sub-set via geological features as appropriate - All RC and aircore sample splitting was to 12.5 % of original sample size or 2 – 3 kg, typical of standard industry practice. Samples greater than 3 kg were split on site before submission to the laboratory. - For core samples, core was separated into sample intervals and separately bagged for analysis at the certified laboratory. - The cyclone and splitter were cleaned every rod string and more frequently when requested by the geologist. In the case of spear sampling for re-splitting purposes, several spears through the entirety of the drill spoil bag were taken in a systematic manner to minimise bias. - Core was cut under the supervision of an experienced geologist, was routinely cut on the orientation line. - Duplicate samples were taken every 20 m from a second cut of the splitter in the case of a cone splitter, or from a reject split in the case of a riffle splitter. Certified standards were inserted into the sample batch at a rate of 1 in 20 throughout all drilling programmes.

Criteria	JORC Code explanation	Commentary
		- Gold at Hall's Creek is fine- to medium-grained and a sample size of 2 – 3 kg is considered appropriate. - Half core is considered appropriate for diamond drill samples.
Quality of assay data and laboratory tests	- The nature, quality and appropriateness of the assaying and laboratory procedures used and whether the technique is considered partial or total. - For geophysical tools, spectrometers, handheld XRF instruments, etc, the parameters used in determining the analysis including instrument make and model, reading times, calibrations factors applied and their derivation, etc. - Nature of quality control procedures adopted (eg standards, blanks, duplicates, external laboratory checks) and whether acceptable levels of accuracy (ie lack of bias) and precision have been established.	- Assays are completed in a certified laboratory in Perth WA - Gold assays are determined using fire assay with 40g charge and AAS finish. Other elements were assayed using acid digest with ICP-MS finish. Screen fire assays consists of screening 500g of the sample to 106 microns. The plus fraction is fire assayed for gold and a duplicate assay is performed on the minus fraction. The size fraction weights, coarse and fine fraction gold content and total gold content are reported. The methods used approach total mineral consumption and are typical of industry standard practice. - Face samples are assayed in the site lab utilising Leachwell bottle roll methodology representing CN recoverable gold. Any samples over 2g/t Au are sent to a certified laboratory in Perth WA lab for confirmation fire assay. All underground face samples prior to March 2017 were fire assayed at an external laboratory. - No geophysical logging of drilling was performed. This is not relevant to the style of mineralisation under exploration. - Lab standards, blanks and repeats are included as part of the QAQC system. In addition the laboratory had its own internal QAQC comprising standards, blanks and duplicates. Sample preparation checks of pulverising at the laboratory include tests to check that the standards of 90% passing 75 micron is being achieved. Follow-up re-assaying is performed by the laboratory upon company request following review of assay data. Acceptable bias and precision is noted in results given the nature of the deposit and the level of classification. QA/QC review on previous drilling shows a negative bias with several of the external certified standards.

Criteria	JORC Code explanation	Commentary
Verification of sampling and assaying	- The verification of significant intersections by either independent or alternative company personnel. - The use of twinned holes. - Documentation of primary data, data entry procedures, data verification, data storage (physical and electronic) protocols. - Discuss any adjustment to assay data.	- Significant intersections are noted in logging and checked with assay results by company personnel. Some significant intersections have been resampled and assayed to validate results. Diamond drilling confirms the width of the mineralised intersections. - The current drill program includes holes testing the current resource and twinning existing RC holes as shown on announcement sections. - All primary data is logged on paper and later entered into the database. Data is visually checked for errors before being sent to an external database manager for further validation and uploaded into an offsite database. Hard copies of original drill logs are kept both onsite and in the Perth office. - No adjustments have been made to assay data.
Location of data points	- Accuracy and quality of surveys used to locate drill holes (collar and down-hole surveys), trenches, mine workings and other locations used in Mineral Resource estimation. - Specification of the grid system used. - Quality and adequacy of topographic control.	- Surface drilling is surveyed using DGPS with an accuracy of $\pm 0.3\text{m}$. Underground Drilling is surveyed using a total station with an accuracy of $\pm 0.2\text{m}$. Downhole surveys are conducted during drilling using single shot cameras at 10 m then every 30 m thereafter. Later drilling was downhole surveyed using a Reflex survey tool. Underground mine workings used in the Mineral Resource estimation are surveyed by company surveyors utilising standard underground survey equipment(Leica jiggers) and established survey controls.Mine workings (open pits) were surveyed by external surveyors using RTK survey equipment. A subset of historical holes was surveyed to validate collar coordinates. - The project lies in MGA 94, zone 52. Local coordinates are derived by conversion: $\text{GDA94_EAST} = \text{NIC_EAST} * 0.9983364 + \text{NIC_NORTH} * 0.05607807 + 315269.176$ $\text{GDA94_NORTH} = \text{NIC_EAST} * (-0.05607807) + \text{NIC_NORTH} * 0.9983364 + 7944798.421$ $\text{GDA94_RL} = \text{NIC_RL} + 2101.799$ - Topographic control uses DGPS collar pickups and external survey RTK data and is considered adequate for use.
Data spacing and distribution	- Data spacing for reporting of Exploration Results. - Whether the data spacing and distribution is sufficient to establish the degree of geological and grade continuity appropriate for the Mineral Resource and Ore Reserve estimation procedure(s) and classifications applied. - Whether sample compositing has been applied.	- Drill hole spacing at Nicolson's is generally between 10 m by 10 m and 30 m x 30 m in the upper areas of the deposits and extends to 40 m x 40 m at depths greater than 200 m. - The Competent Person is of the view that the drill spacing, geological interpretation and grade continuity of the data supports the resource categories assigned. - Where used historically sample compositing to 3m occurred in holes above predicted mineralized zones. Composite samples were re-assayed in their 1 m increments if initial assay results were anomalous.

Criteria	JORC Code explanation	Commentary
Orientation of data in relation to geological structure	- Whether the orientation of sampling achieves unbiased sampling of possible structures and the extent to which this is known, considering the deposit type. - If the relationship between the drilling orientation and the orientation of key mineralised structures is considered to have introduced a sampling bias, this should be assessed and reported if material.	- Surface Drilling is predominantly at 270o to local grid at a dip of -60o. Local structures strike north-south on the local grid and dip at 60oE. No bias of sampling is believed to exist through the drilling orientation - Underground development sampling is nominally undertaken normal to the various orebodies. - Underground drill holes are designed to drill across geological structures i.e. not along geological structures.
Sample security	The measures taken to ensure sample security.	The chain of custody is managed by Pantoro employees and consultants. Samples are stored on site and delivered in sealed boxes and bags to the lab in Perth. Samples are tracked during shipping. Samples are reconciled at the assay lab.
Audits or reviews	The results of any audits or reviews of sampling techniques and data.	- A review of the resource was carried out by an independent consultancy firm when the project was acquired from Bulletin. No significant issues were noted. - A review of the historic sampling techniques was carried out by an independent consultancy in relation to prior Mineral Resource estimation for Bulletin Resources in 2011/12 on behalf of the previous owners. No significant issues were noted in the 2007-2011 dataset.

SECTION 2: REPORTING OF EXPLORATION RESULTS – NICOLSONS

Criteria	JORC Code explanation	Commentary
Mineral tenement and land tenure status	- Type, reference name/number, location and ownership including agreements or material issues with third parties such as joint ventures, partnerships, overriding royalties, native title interests, historical sites, wilderness or national park and environmental settings. - The security of the tenure held at the time of reporting along with any known impediments to obtaining a licence to operate in the area.	- Tenements containing Mineral Resources and Ore Reserves are 100% held by Pantoro subsidiary company Halls Creek Mining Pty Ltd. This is: M80/359. Tenement transfers to HCM are yet to occur as stamp duty assessments have not been completed by the office of state revenue. The tenements lie on a pastoral lease with access and mining agreements and predate native title claims. - The tenements are in good standing and no known impediments exist.
Exploration done by other parties	Acknowledgment and appraisal of exploration by other parties.	The deposits were discovered by prospectors in the early 1990s. After an 8,500 m RC program, Precious Metals Australia mined 23 koz at an estimated 7.7g/t Au from Nicolson's Pit in 1995/96 before ceasing the operation. Rewah mined the Wagtail and Rowdy pits (5 koz at 2.7g/t Au) in 2002/3 before Terra Gold Mines (TGM) acquired the project, carried out 12,000 m of RC drilling and produced a 100 koz Mineral Resource estimate for the Nicolson's Find deposit. GBS Gold acquired TGM and drilled 4,000 m before being placed in administration. Bulletin Resources Ltd acquired the project from administrators and conducted exploration work focused on Nicolson's and the Wagtail Deposits and completed regional exploration drilling and evaluation and completed a Mining Study in 2012 which included Mineral Resource and Ore Reserves completed by independent consultants prior to entering into a JV with PNR in 2014. Review of available reports show work to follow acceptable to standard industry practices.

Criteria	JORC Code explanation	Commentary
Geology	Deposit type, geological setting and style of mineralisation.	- Gold mineralisation in the Nicolson's Find area is structurally controlled within the 400 m wide NNE trending dextral strike slip Nicolson's Find Shear Zone (NFSZ) and is hosted within folded and metamorphosed turbiditic greywackes, felsic volcanoclastics, mafic volcanics and laminated siltstones and mudstones. This zone forms part of a regional NE-trending strike slip fault system developed across the Halls Creek Orogen (HCO). - The NFSZ comprises a NNE-trending anastomosing system of brittle-ductile shears, characterised by a predominantly dextral sense of movement. The principal shear structures trend NNE to N-S and are linked by NW, and to a lesser extent, by NE shears. Individual shears extend up to 800m along strike and overprint the earlier folding and penetrative cleavage of the HCO. - The overall geometry of the system is characterized by right step-overs and bends/jogs in the shear traces, re-flecting refraction of the shears about the granite contact. Within this system, the NW-striking shears are interpreted as compressional structures and the NE-striking shears formed within extensional windows. - Mineralisation is primarily focussed along NNE trending anastomosing systems of NNE-SSW, NW-SE and NE-SW oriented shears and splays. The NNE shears dip moderately to the east, while the NW set dips moderately to steeply to the NE. Both sets display variations in dip, with flattening and steepening which result in a complex pattern of shear intersections.. - Mineralisation is strongly correlated with discontinuous quartz veining and with Fe-Si-K alteration halos developed in the wall rocks to the veins. The NE shears are associated with broad zones of silicification and thicker quartz veining (typically white, massive quartz with less fracturing and brecciation); however, these are typically poorly mineralized. The NW-trending shears are mineralized, with the lodes most likely related to high fluid pressures with over-pressuring and failure leading to vein formation. Although the NE structures formed within the same shear system, the quartz veining is of a different generation to the mineralized veins. - Individual shears within the system display an increase in strain towards their centres and comprise an anastomosing shear fabric reminiscent of the pattern on a larger scale.

Criteria	JORC Code explanation	Commentary
Drill hole Information	- A summary of all information material to the understanding of the exploration results including a tabulation of the following information for all Material drill holes: » easting and northing of the drill hole collar » elevation or RL (Reduced Level – elevation above sea level in metres) of the drill hole collar » dip and azimuth of the hole » down hole length and interception depth » hole length. - If the exclusion of this information is justified on the basis that the information is not Material and this exclusion does not detract from the understanding of the report, the Competent Person should clearly explain why this is the case.	No exploration results are reported as part of this release, results relating to the deposits have been previously released.
Data aggregation methods	- In reporting Exploration Results, weighting averaging techniques, maximum and/or minimum grade truncations (eg cutting of high grades) and cut-off grades are usually Material and should be stated. - Where aggregate intercepts incorporate short lengths of high grade results and longer lengths of low grade results, the procedure used for such aggregation should be stated and some typical examples of such aggregations should be shown in detail. - The assumptions used for any reporting of metal equivalent values should be clearly stated.	No exploration results are reported as part of this release, results relating to the deposit have been previously released.
Relationship between mineralisation widths and intercept lengths	- These relationships are particularly important in the reporting of Exploration Results. - If the geometry of the mineralisation with respect to the drill hole angle is known, its nature should be reported. - If it is not known and only the down hole lengths are reported, there should be a clear statement to this effect (eg 'down hole length, true width not known').	- Drilling is predominantly at 270o to local grid at a dip of -60o. Local structures strike 0o to the local grid and dip at 60oE (i.e. having a 60o intersection angle to lode structures). Deeper holes have some drill hole deviation which decreases or increases the intersection angle, but not to a significant extent. - Face mapping data supports widths interpreted from drill holes - Downhole lengths are reported and true widths are approximately 60 – 90% of down-hole length. True widths are calculated and reported for any drill intersections > 1 ppm Au.
Diagrams	Appropriate maps and sections (with scales) and tabulations of intercepts should be included for any significant discovery being reported These should include, but not be limited to a plan view of drill hole collar locations and appropriate sectional views.	No exploration results are reported as part of this release, and therefore no diagrams are included.
Balanced reporting	Where comprehensive reporting of all Exploration Results is not practicable, representative reporting of both low and high grades and/or widths should be practiced to avoid misleading reporting of Exploration Results.	Results greater than 1 ppm Au have been previously reported for the recent drilling.

Criteria	JORC Code explanation	Commentary
Other substantive exploration data	Other exploration data, if meaningful and material, should be reported including (but not limited to): geological observations; geophysical survey results; geochemical survey results; bulk samples – size and method of treatment; metallurgical test results; bulk density, groundwater, geotechnical and rock characteristics; potential deleterious or contaminating substances.	No exploration results are reported as part of this release
Further work	- The nature and scale of planned further work (eg tests for lateral extensions or depth extensions or large-scale step-out drilling). - Diagrams clearly highlighting the areas of possible extensions, including the main geological interpretations and future drilling areas, provided this information is not commercially sensitive.	Underground diamond drilling is ongoing on a continuous shift basis and will continue to test for the extension of the deposit which remains open.

SECTION 3: ESTIMATION AND REPORTING OF MINERAL RESOURCES – NICOLSONS

Criteria	JORC Code explanation	Commentary
Database integrity	- Measures taken to ensure that data has not been corrupted by, for example, transcription or keying errors, between its initial collection and its use for Mineral Resource estimation purposes. - Data validation procedures used.	- Data input has been governed by lookup tables and programmed import of assay data from lab into database. The database has been checked against the original assay certificates and survey records for completeness and accuracy. - Data was validated by the geologist after input. Data validation checks were carried out by an external database manager in liaison with Pantoro personnel. The database was further validated by external resource consultants prior to resource modelling. An extensive review of the data base was undertaken when Pantoro acquired the project, and external data review is ongoing.
Site visits	- Comment on any site visits undertaken by the Competent Person and the outcome of those visits. - If no site visits have been undertaken indicate why this is the case.	The Competent Person regularly visits the site and has a good appreciation of the mineralisation styles comprising the Mineral Resource.
Geological interpretation	- Confidence in (or conversely, the uncertainty of) the geological interpretation of the mineral deposit. - Nature of the data used and of any assumptions made. - The effect, if any, of alternative interpretations on Mineral Resource estimation. - The use of geology in guiding and controlling Mineral Resource estimation. - The factors affecting continuity both of grade and geology.	- Confidence in the geological interpretation is generally proportional to the drill density. Surface mapping confirms some of the orientation data for the main mineralised structures. - Data used for the geological interpretation includes surface and trench mapping and drill logging data. Underground face sampling, face geology and backs mapping were also utilized from close spaced level development is also used where available. - In general, the interpretation of the mineralised structures is clear. - Geological interpretation of the data was used as a basis for the lodes which were then constrained by cut-off grades. - Geology and grade continuity are constrained by quartz veining within the NFSZ and by parallel structures for the other prospects.
Dimensions	The extent and variability of the Mineral Resource expressed as length (along strike or otherwise), plan width, and depth below surface to the upper and lower limits of the Mineral Resource.	The Nicolson's deposit is approximately 950 m in strike length and generally 0.5 to 2m wide and extends nominally 500 m metres below surface.

Criteria	JORC Code explanation	Commentary
Estimation and modelling techniques	- The nature and appropriateness of the estimation technique(s) applied and key assumptions, including treatment of extreme grade values, domaining, interpolation parameters and maximum distance of extrapolation from data points. If a computer assisted estimation method was chosen include a description of computer software and parameters used. - The availability of check estimates, previous estimates and/or mine production records and whether the Mineral Resource estimate takes appropriate account of such data. - The assumptions made regarding recovery of by-products. - Estimation of deleterious elements or other non-grade variables of economic significance (e.g. sulphur for acid mine drainage characterisation). - In the case of block model interpolation, the block size in relation to the average sample spacing and the search employed. - Any assumptions behind modelling of selective mining units. - Any assumptions about correlation between variables. - Description of how the geological interpretation was used to control the resource estimates. - Discussion of basis for using or not using grade cutting or capping. - The process of validation, the checking process used, the comparison of model data to drill hole data, and use of reconciliation data if available.	- A block model was generated for the Nicolson's deposit. Individual mineralised structures were domained separately. Models contain grade estimates and attributes for blocks within each domain only. - Two domains were updated during the 2019 Nicolson's MRE, these being Domain 4 (Johnston Lode) and Domain 49 (Anderson Lode), all other domains remain as per the 2018 Nicolson's MRE. - Geological interpretation forms the basis for the mineralisation domain wireframes; these were oriented along trends of grade continuity and form hard boundaries during estimation. - A two-dimensional ("2D") Ordinary Kriging (OK) interpolation approach was selected to address some of the main issues encountered when estimating narrow vein mineralisation, such as: - Additivity issues due to non-uniform support and resulting grade bias. Instances of highly variable individual intercepts (e.g. 0.3 m to 5.0 m) which would be difficult to incorporate and represent statistically using downhole composites of equal lengths (e.g. 0.5, 1.0 or 2.0 m); - Varying mineralisation geometry across lode, down dip, and along strike; and - Block size required for adequate volume fill of narrow geometry is generally too small, introducing conditional bias to the MRE outcome. - Drillholes were composited for the full width of the domain intercept, followed by trigonometric calculation of true width ("TW") using the orientations of the drill hole intercept and ore domain defined by a digitized the Leapfrog reference (centerline) surface. A gold accumulation variable was then calculated by multiplication of intercept grade by true width. - Composited sample data was pressed onto a cartographic plane and statistical analysis undertaken on accumulation, width, and grade variables, to assist with determining estimation search parameters, top cuts etc. - Assessment and application of top-cutting for the 2D estimate was undertaken on the gold accumulation variable within individual domains. Top cuts, where appropriate, were applied on an individual domain basis. - Top cuts were applied to the gram-meter accumulation variable after statistical, spatial analysis and assessment of percentage of metal reduction within each mineralized domain with cut values being: - Domain 4 = 160 g/m Accumulation and a 6.5% metal reduction, - Domain 49 = 160 g/m Accumulation and a 11.1% metal reduction. - Variography analysis of individual domains was undertaken on gold accumulation variables in 2D space, followed by Qualitative Kriging Neighbourhood Analysis to assist with determining appropriate search parameters.

Criteria	JORC Code explanation	Commentary
Estimation and modelling techniques		- The 2D block models for interpolation were created using a block size of 10 mN x 10 mRL x 1 mE with no sub-celling. Block size was determined primarily with the assumption of a relatively selective mining approach for underground operations. - The search strategy was a maximum extrapolation distance of 120 m and 135 m over three search passes for Domains 4 and 49 respectively. The first pass search was equal to the variogram maximum range (40 m for Domain 4 and 45 m for Domain 49) with the second pass search double the variogram range (80 m for Domain 4 and 90 m for Domain 49) and the third pass triple the variogram range (120 m for Domain 4 and 135 m for Domain 49). A constant minimum of 4 and maximum of 12 composites was maintained across all search passes for Domain 4. Domain 49 utilised a minimum of 4 and a maximum of 12 for the first pass, dropping to a minimum of 3 samples for the second and third passes.. - A distance based high-grade limit function was applied to Domain 49, limiting accumulation values above 60 gram-metres to half the variogram range (22.5 m). - Post estimate. Gold ppm values for each block were calculated by dividing interpolated gold accumulation by interpolated TW, whereby for each block: - Block Gold ppm = Block Gold Accumulation Value / Block TW Value - Back calculated gold ppm values for each block were transformed from 2D to 3D space and pressed across the full width of the corresponding domain in the final host 3D compilation model. - Check estimates for both domains were carried out in 2D using Inverse Distance Squared. Both accumulation and true width were estimated before back calculation of the check estimate gold grade. - Validation of the gold accumulation, TW estimations and gold ppm back-calculation was completed by global and local bias analysis, statistical and visual inspections in 2D and 3D space. - By products are not included in the resource estimate. - No deleterious elements have been estimated. Arsenic is known to be present, however metallurgical test work suggests that it does not adversely affect metallurgical recovery.
Moisture	Whether the tonnages are estimated on a dry basis or with natural moisture, and the method of determination of the moisture content	- Tonnage was estimated on a dry basis. - The tonnages of material on stockpiles are quoted on a dry basis.
Cut-off parameters	The basis of the adopted cut-off grade(s) or quality parameters applied	The global gold Mineral Resource has been reported at a 2.5 g/t gold cut-off for Underground (greater than 100 m below topographic surface) resources and a 0.6 g/t cutoff for Open Pit (within 100 m of topographic surface) resources and is based upon economic parameters currently utilised at the Nicolson operations, where deposits of the same style, commodity, comparable size and mining methodology are currently being extracted.

Criteria	JORC Code explanation	Commentary
Mining factors or assumptions	Assumptions made regarding possible mining methods, minimum mining dimensions and internal (or, if applicable, external) mining dilution. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider potential mining methods, but the assumptions made regarding mining methods and parameters when estimating Mineral Resources may not always be rigorous. Where this is the case, this should be reported with an explanation of the basis of the mining assumptions made.	The MRE extends nominally 500 m below topographic surface and lies within 100 vertical metres of active level development. Pantoro considers material at this depth would fall within the definition of 'reasonable prospect of eventual economic extraction' within an underground mining framework.
Metallurgical factors or assumptions	The basis for assumptions or predictions regarding metallurgical amenability. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider potential metallurgical methods, but the assumptions regarding metallurgical treatment processes and parameters made when reporting Mineral Resources may not always be rigorous. Where this is the case, this should be reported with an explanation of the basis of the metallurgical assumptions made.	Metallurgical test work has shown acceptable (> 92%) gold recovery using CIP technology and is consistent with calculated recoveries from the current operating period from the Nicolson's underground mine. No factors from the metallurgy have been applied to the estimates.
Environmental factors or assumptions	Assumptions made regarding possible waste and process residue disposal options. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider the potential environmental impacts of the mining and processing operation. While at this stage the determination of potential environmental impacts, particularly for a greenfields project, may not always be well advanced, the status of early consideration of these potential environmental impacts should be reported. Where these aspects have not been considered this should be reported with an explanation of the environmental assumptions made.	The deposits are on granted mining leases with existing mining disturbance and infrastructure present.
Bulk density	- Whether assumed or determined. If assumed, the basis for the assumptions. If determined, the method used, whether wet or dry, the frequency of the measurements, the nature, size and representativeness of the samples. - The bulk density for bulk material must have been measured by methods that adequately account for void spaces (vugs, porosity, etc.), moisture and differences between rock and alteration zones within the deposit. - Discuss assumptions for bulk density estimates used in the evaluation process of the different materials.	- Bulk density measurements of ore were calculated from drill core and underground samples using the water displacement method and data from historical mining. - Bulk densities vary due to ore type and are assigned separately to each domain based on this work.

Criteria	JORC Code explanation	Commentary
Classification	- The basis for the classification of the Mineral Resources into varying confidence categories. - Whether appropriate account has been taken of all relevant factors (i.e. relative confidence in tonnage/grade estimations, reliability of input data, confidence in continuity of geology and metal values, quality, quantity and distribution of the data). - Whether the result appropriately reflects the Competent Person's view of the deposit.	- This Mineral Resource Estimate has been classified as Measured, Indicated and Inferred to appropriately represent confidence and risk with respect to data quality, drill hole spacing, geological and grade continuity, mineralisation volumes, recent and historical mining activity as well as metal distribution. Additional considerations were the stage of project assessment, amount of diamond drilling, current understanding of mineralisation controls and selectivity within an underground mining environment. The Nicolson's deposit has been mined continuously by Underground methods since mid-2015 with recent data from underground production supporting both grade and geological continuity. The bulk of the data utilised in the current Mineral Resource estimate is from recently acquired drilling and sampling with an additional 27,042 m drilling from 256 reverse circulation and diamond holes as well as 6,674 m of sampling from 2,047 underground production faces. - This approach considers all relevant factors and reflects the Competent Person's view of the deposit.
Audits or reviews	The results of any audits or reviews of Mineral Resource estimates	The current Mineral Resource has been reviewed internally and results are consistent with reconciled production results.
Discussion of relative accuracy/ confidence	- Where appropriate a statement of the relative accuracy and confidence level in the Mineral Resource estimate using an approach or procedure deemed appropriate by the Competent Person. For example, the application of statistical or geostatistical procedures to quantify the relative accuracy of the resource within stated confidence limits, or, if such an approach is not deemed appropriate, a qualitative discussion of the factors that could affect the relative accuracy and confidence of the estimate. - The statement should specify whether it relates to global or local estimates, and, if local, state the relevant tonnages, which should be relevant to technical and economic evaluation. Documentation should include assumptions made and the procedures used. - These statements of relative accuracy and confidence of the estimate should be compared with production data, where available.	- The relative accuracy of the Mineral Resource estimate is reflected in the reporting of the Mineral Resource as per the guidelines of the 2012 JORC Code. - The statement reflects a global estimate of tonnes and grade. - The current Mineral Resource estimate produced a 5% oz Au undercall against mine claim production for the Nicolson's underground mined to date. This amount is considered to be within acceptable limits for the classification of the Mineral Resource.

SECTION 4: ESTIMATION AND REPORTING OF ORE RESERVES – NICOLSONS

Criteria	JORC Code explanation	Commentary
Mineral Resource estimate for conversion to Ore Reserves	- Description of the Mineral Resource estimate used as a basis for the conversion to an Ore Reserve. - Clear statement as to whether the Mineral Resources are reported additional to, or inclusive of, the Ore Reserves.	- The Ore Reserve was calculated using detailed mine designs applied to the current Mineral Resource estimate. The Mineral Resource estimate was completed by experienced geologists familiar with the deposits, overseen by the competent person. - The Mineral Resources reported are inclusive of the Ore Reserve.
Site visits	- Comment on any site visits undertaken by the Competent Person and the outcome of those visits. - If no site visits have been undertaken indicate why this is the case.	The Competent Person makes regular visits to the site and is involved in preparation of the overall operations plans which are the basis for the Ore Reserve.
Study status	- The type and level of study undertaken to enable Mineral Resources to be converted to Ore Reserves. - The Code requires that a study to at least Pre-Feasibility Study level has been undertaken to convert Mineral Resources to Ore Reserves. Such studies will have been carried out and will have determined a mine plan that is technically achievable and economically viable, and that material Modifying Factors have been considered.	- Mine designs were completed using the Mineral Resource as the basis for the conversion to Ore Reserves and is considered to be at an operational level. - Cut off grades were relevant to actual costs at the operation. - Modifying Factors applicable to results from current operations are utilised. - The mine is currently operating on a profitable basis and cut off grades were chosen according to actual costs. Low grade material was included in the Ore Reserve when required to be mined as a result of mining method or Ore geometry.
Cut-off parameters	The basis of the cut-off grade(s) or quality parameters applied.	- Nicolsons - The fully costed cut off grade is approximately 4.4 g/t. Utilising the current resale and cut and fill mining methods, contiguous blocks of ore are required to be mined, and material with a grade above 2.4g/t was included in the Ore Reserve where required to facilitate mining of the high grade ore. Material with lower grades is assumed to be discarded as waste. - Nicolsons open pit – Pits were designed using a 2.14g/t cut-off grade. Low grade resources recovered below the cut grade have not been reported, but in practice would be report to low grade stockpiles.

Criteria	JORC Code explanation	Commentary
Mining factors or assumptions	- The method and assumptions used as reported in the Pre-Feasibility or Feasibility Study to convert the Mineral Resource to an Ore Reserve (i.e. either by application of appropriate factors by optimisation or by preliminary or detailed design). - The choice, nature and appropriateness of the selected mining method(s) and other mining parameters including associated design issues such as pre-strip, access, etc. - The assumptions made regarding geotechnical parameters (eg pit slopes, stope sizes, etc), grade control and pre-production drilling. - The major assumptions made and Mineral Resource model used for pit and stope optimisation (if appropriate). - The mining dilution factors used. - The mining recovery factors used. - Any minimum mining widths used. - The manner in which Inferred Mineral Resources are utilised in mining studies and the sensitivity of the outcome to their inclusion. - The infrastructure requirements of the selected mining methods.	Nicolsons Underground - For Development and benching of development floors, ore was assumed to be resue mined, with a minimum resue ore width of 1.65.0m. When the Ore was modelled at less than minimum width in the mineral Resource, grade was diluted by waste material at zero grade to achieve minimum width ("Insitu Grade"). The Insitu Grade was diluted by a further 25% dilution which is consistent with current development experience at Nicolsons. Ore recovery is 100% in ore drives and benches. - For resue stoping and cut and fill stoping a minimum width of 1.65m was applied, with waste dilution at zero grade applied where the ore zone is narrower than the minimum width. The Insitu Grade was diluted by a further 25% dilution which is consistent with current mining practice at Nicolsons. Ore recovery is 95% for resue and cut and fill stopes. - For air leg stoping, a minimum width of 2m was applied. 25% dilution and 95% recovery was applied. - For the Nicolsons underground Ore Reserve, Ore is extracted using a number of mining methods. The majority of the Ore is extracted using resue mining or cut and fill methods. Resue mining involves mining an uphole stope approximately 4m in height along the entire length of the developed ore block at the minimum grade or wider. Once the ore is extracted, the remaining waste is stripped to full drive width. This process is repeated, working off of the waste blasted into the drive below (and additional introduced waste if required). The final resue stope is left open beneath a cemented fill pillar. Cut and fill stoping involves filling the ore drive before ramping up onto the waste fill and stripping a new drive sized excavation with Ore blasted and bogged before the waste is blasted. Air leg stopes are mined as leading stopes between levels. - Minimum mining widths, dilution and recovery assumptions are stipulated above. - Mining is by owner operator using leased equipment. Actual lease rates and manning costs are utilised. - Inferred Mineral Resources are excluded from the Ore Reserve. The Ore Reserve is considered feasible without the inclusion of Inferred Mineral Resources. - The costs used in the Ore Reserve include all required infrastructure including fixed plant, buildings and magazines, and mine excavations.

Criteria	JORC Code explanation	Commentary
		Open Pits - Mineral Resources were optimized using whittle 4D software using A\$1,400/oz gold price, followed by detailed open pit design using Surpac software. - Key parameters used in optimisation were sourced from prevailing site prices (fuel and consumables, milling cost and administration cost), contract rates (mining) and prevailing market rates for general items. - Final overall pit slopes are 43 degrees, in line with geotechnical recommendation's by the geotechnical consultant. - Mining dilution of 15% and 100% recovery of diluted ore was utilised.
Metallurgical factors or assumptions	- The metallurgical process proposed and the appropriateness of that process to the style of mineralisation. - Whether the metallurgical process is well-tested technology or novel in nature. - The nature, amount and representativeness of metallurgical test work undertaken, the nature of the metallurgical domaining applied and the corresponding metallurgical recovery factors applied. - Any assumptions or allowances made for deleterious elements. - The existence of any bulk sample or pilot scale test work and the degree to which such samples are considered representative of the orebody as a whole. - For minerals that are defined by a specification, has the ore reserve estimation been based on the appropriate mineralogy to meet the specifications?	- The existing processing plant at Nicolsons uses a conventional CIP circuit, which is appropriate for the style of mineralization, and has achieved approximately 92% recovery during the past year. - The CIP process is the conventional gold processing method in Western Australia and is well tested and proven. - The site is an operating mine with recovery of 92% at usual operating condition. The site has undertaken ongoing testing of new ore samples, with similar results achieved in the laboratory. - There are not any know deleterious elements - The 92% recovery is consistent with calculated recoveries from the current operating period from the Nicolsons underground mine - Not applicable - A Steinert multi-sensor ore sorter is installed at the project. While the ore sorter is beneficial to site production and lower production costs, the benefits of the ore sorter use have not been factored into the Ore Reserve calculation.
Infrastructure	The existence of appropriate infrastructure: availability of land for plant development, power, water, transportation (particularly for bulk commodities), labour, accommodation; or the ease with which the infrastructure can be provided, or accessed.	The Nicolsons site has been in production since September 2015, and all infrastructure and services necessary to operate the mine are in place and functioning.

Criteria	JORC Code explanation	Commentary
Costs	- The derivation of, or assumptions made, regarding projected capital costs in the study - The methodology used to estimate operating costs. - Allowances made for the content of deleterious elements. - The source of exchange rates used in the study. - Derivation of transportation charges. - The basis for forecasting or source of treatment and refining charges, penalties for failure to meet specification, etc. - The allowances made for royalties payable, both Government and private.	- Major project capital for the project is already in place at the mine. Ongoing sustaining capital is identified in site budgets and scheduled as required. - Operating costs are calculated using a combination of actual unit costs at the mine and first as appropriate. Actual labour costs are utilized in site budgets. - There are no known deleterious elements and no adjustments have been made. - All costs were estimated in Australian dollars, and a gold price of \$1800/Oz was assumed. - Transport charges were based on actual costs during the past year. - The 2.5% state government royalty was included in the detailed budget. No other royalties are applicable to the project.
Revenue factors	- The derivation of, or assumptions made regarding revenue factors including head grade, metal or commodity price(s) exchange rates, transportation and treatment charges, penalties, net smelter returns, etc. - The derivation of assumptions made of metal or commodity price(s), for the principal metals, minerals and co-products.	- Production, Grade, and cost is scheduled monthly in a detailed operations budget schedule. - Gold price was assumed to be A\$1,800 per ounce. - No revenue assumed for silver production, however in practice historically approximately 1 oz of silver is produced and sold for every 2.5 oz of gold.
Market assessment	- The demand, supply and stock situation for the particular commodity, consumption trends and factors likely to affect supply and demand into the future. - A customer and competitor analysis along with the identification of likely market windows for the product. - Price and volume forecasts and the basis for these forecasts. - For industrial minerals the customer specification, testing and acceptance requirements prior to a supply contract.	- Gold prices can be volatile and there are many conflicting positions on the future price of Gold. Pantoro believes that A\$1,800 per ounce is a realistic forward price forecast for gold over the life of the proposed mine. Pantoro holds a number of gold hedge positions to April 2020 with the average hedge price of \$1735/Oz for September and October 2019 and \$1800/Oz for November 2019 – April 2020. - Gold is sold on the spot market and customer/competitor analysis is unnecessary.
Economic	- The inputs to the economic analysis to produce the net present value (NPV) in the study, the source and confidence of these economic inputs including estimated inflation, discount rate, etc. - NPV ranges and sensitivity to variations in the significant assumptions and inputs.	- As the mine is in operation with a relatively short mine life and all major capital cost elements already in place, an NPV analysis was not required. - Ore grade, gold price and site costs are the key sensitivities for the project.
Social	The status of agreements with key stakeholders and matters leading to social licence to operate.	The Ore Reserve is located on granted mining leases and the company has an access agreement with the pastoral lease owner who is also the local aboriginal corporation.

Criteria	JORC Code explanation	Commentary
Other	- To the extent relevant, the impact of the following on the project and/or on the estimation and classification of the Ore Reserves: - Any identified material naturally occurring risks. - The status of material legal agreements and marketing arrangements. - The status of governmental agreements and approvals critical to the viability of the project, such as mineral tenement status, and government and statutory approvals. There must be reasonable grounds to expect that all necessary Government approvals will be received within the timeframes anticipated in the Pre-Feasibility or Feasibility study. Highlight and discuss the materiality of any unresolved matter that is dependent on a third party on which extraction of the reserve is contingent.	- Pantoro is the 100% owner of the Project. Pantoro is not currently named on tenement titles due to delays in assessment of stamp duty by the Office of State Revenue. Pantoro is satisfied that it has full legal rights to 100% of the project by virtue of the legal agreements in place with the previous owner. - Signed transfer documents for the tenements have been lodged with MIRS for 84% if the tenement interest and the transfers are expected to be completed in the near term. The office of state revenue is yet to provide an assessment for the final 16%.. - PNR has the required government and stakeholder approvals required to mine and process the Ore Reserve
Classification	- The basis for the classification of the Ore Reserves into varying confidence categories. - Whether the result appropriately reflects the Competent Person's view of the deposit. - The proportion of Probable Ore Reserves that have been derived from Measured Mineral Resources (if any).	- The Ore has been derived from Measured and Indicated Resources. Inferred Mineral Resource has been excluded from the Ore Reserve. - Proven Ore Reserves are derived from Measured Mineral Resources. - Probable Ore Reserves are derived from Indicated Mineral Resources. - The Competent Person has been closely involved in operations and planning at the mine since commencement in 2015. The Competent Person is satisfied the Ore Reserve reflects the actual results from the operation.
Audits or reviews	The results of any audits or reviews of Ore Reserve estimates.	The Ore Reserve has been internally audited by Geologists and Mining Engineers involved with operations at the mine. No external audit or review has been undertaken

Criteria	JORC Code explanation	Commentary
Discussion of relative accuracy/ confidence	- Where appropriate a statement of the relative accuracy and confidence level in the Ore Reserve estimate using an approach or procedure deemed appropriate by the Competent Person. For example, the application of statistical or geostatistical procedures to quantify the relative accuracy of the reserve within stated confidence limits, or, if such an approach is not deemed appropriate, a qualitative discussion of the factors which could affect the relative accuracy and confidence of the estimate. - The statement should specify whether it relates to global or local estimates, and, if local, state the relevant tonnages, which should be relevant to technical and economic evaluation. Documentation should include assumptions made and the procedures used. - Accuracy and confidence discussions should extend to specific discussions of any applied Modifying Factors that may have a material impact on Ore Reserve viability, or for which there are remaining areas of uncertainty at the current study stage. - It is recognised that this may not be possible or appropriate in all circumstances. These statements of relative accuracy and confidence of the estimate should be compared with production data, where available.	- The Proven Ore Reserve is primarily based on close spaced underground development sampling and mapping on the currently developed areas of the mine - The Probable Ore Reserve is primarily based on a combination of Diamond and RC drilling. Diamond drilling and mine development indicates that ore is generally narrower, but higher grade than indicated by RC drilling. Reliance on RC drilling is decreasing as the mine progresses lower. - The Ore Reserve is noted to be consistent in grade and nature compared with historical operations at the mine. This assessment was undertaken by way of a general review and is not based on statistical analysis. - No Modifying Factors apart from those set out in this Table 1 have been included. Metallurgical recovery factors have not been applied to the Ore Reserve. - Stopes were designed using horizontal and vertical development shapes actually designed for production. The designs reflect current mine planning practice at the mine.

APPENDIX B – JORC CODE 2012 EDITION – TABLE 1

SECTION 1: SAMPLING TECHNIQUES AND DATA – GRANTS CREEK

Criteria	JORC Code explanation	Commentary
Sampling techniques	- Nature and quality of sampling (eg cut channels, random chips, or specific specialised industry standard measurement tools appropriate to the minerals under investigation, such as down hole gamma sondes, or handheld XRF instruments, etc). These examples should not be taken as limiting the broad meaning of sampling. - Include reference to measures taken to ensure sample representivity and the appropriate calibration of any measurement tools or systems used. - Aspects of the determination of mineralisation that are Material to the Public Report. - In cases where 'industry standard' work has been done this would be relatively simple (eg 'reverse circulation drilling was used to obtain 1 m samples from which 3 kg was pulverised to produce a 30 g charge for fire assay'). In other cases more explanation may be required, such as where there is coarse gold that has inherent sampling problems. Unusual commodities or mineralisation types (eg submarine nodules) may warrant disclosure of detailed information.	- This information in this release relates to a maiden Mineral Resource estimate for the Perseverance and Star of Kimberley prospect at the Grants Creek gold project. - RC – Rig-mounted static splitter used, with sample falling through a riffle splitter, splitting the sample in 87.5/12.5 ratio sampled every 1m - RC samples 2-5kg samples are dispatched to an external accredited laboratory (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). - Diamond samples 2-5kg samples are dispatched to an external accredited laboratory (BVA Perth) where they are crushed and pulverized to a pulp (P90 75 micron) for fire assay (40g charge). - All core is logged and sampled according to geology, with only selected samples assayed. Core is halved, with RHS of cutting line assayed, and the other half retained in core trays on site for further analysis. Samples are a maximum of 1.2m, with shorter intervals utilised according to geology to a minimum interval of 15m where clearly defined mineralisation is evident. - Core is aligned, measured and marked up in metre intervals referenced back to downhole core blocks .
Drilling techniques	Drill type (eg core, reverse circulation, open-hole hammer, rotary air blast, auger, Bangka, sonic, etc) and details (eg core diameter, triple or standard tube, depth of diamond tails, face-sampling bit or other type, whether core is oriented and if so, by what method, etc).	- RC – Reverse circulation drilling was carried out using a face sampling hammer and a 130mm diameter bit - Surface DD – HQ and NQ2 diamond tails completed on 3m rock roller pre-collars, all core has orientations completed
Drill sample recovery	- Method of recording and assessing core and chip sample recoveries and results assessed. - Measures taken to maximise sample recovery and ensure representative nature of the samples. - Whether a relationship exists between sample recovery and grade and whether sample bias may have occurred due to preferential loss/gain of fine/coarse material.	- All holes were logged at site by an experienced geologist. Recovery and sample quality were visually observed and weights recorded at the laboratory - RC- recoveries are monitored by visual inspection of split reject and lab weight samples are recorded and reviewed. - DD – No significant core loss has been noted in fresh material. Good core recovery has generally been achieved in all sample types in the current drilling program.
Logging	- Whether core and chip samples have been geologically and geotechnically logged to a level of detail to support appropriate Mineral Resource estimation, mining studies and metallurgical studies. - Whether logging is qualitative or quantitative in nature. Core (or costean, channel, etc) photography. - The total length and percentage of the relevant intersections logged.	- Geological logging is completed by a qualified geologist and logging parameters include: depth from, depth to, condition, weathering, oxidation, lithology, texture, colour, alteration style, alteration intensity, alteration mineralogy, sulphide content and composition, quartz content, veining, and general comments. 100% of the drill holes are logged geologically

Criteria	JORC Code explanation	Commentary
Sub-sampling techniques and sample preparation	- If core, whether cut or sawn and whether quarter, half or all core taken. - If non-core, whether riffled, tube sampled, rotary split, etc and whether sampled wet or dry. - For all sample types, the nature, quality and appropriateness of the sample preparation technique. - Quality control procedures adopted for all sub-sampling stages to maximise representivity of samples. - Measures taken to ensure that the sampling is representative of the in situ material collected, including for instance results for field duplicate/second-half sampling. - Whether sample sizes are appropriate to the grain size of the material being sampled.	- All RC holes are sampled on 1m intervals. - RC samples are taken off the rig splitter, no significant water is encountered and are typically dry - Core samples were sawn in half utilising an Almonte core-saw, with RHS of cutting line sent for assaying and the other half retained in core trays on site for future analysis. - For core samples, core was separated into sample intervals and separately bagged for analysis at the certified laboratory. - Core was cut under the supervision of an experienced geologist, it was routinely cut on the orientation line. - All mineralised zones are sampled as well as material considered barren either side of the mineralised interval - Field duplicates for RC samples were taken as part of this program. - Half core is considered appropriate for diamond drill samples. - Sample sizes are considered appropriate for the material being sampled and weights are recorded and monitored by project geologists.
Quality of assay data and laboratory tests	- The nature, quality and appropriateness of the assaying and laboratory procedures used and whether the technique is considered partial or total. - For geophysical tools, spectrometers, handheld XRF instruments, etc, the parameters used in determining the analysis including instrument make and model, reading times, calibrations factors applied and their derivation, etc. - Nature of quality control procedures adopted (eg standards, blanks, duplicates, external laboratory checks) and whether acceptable levels of accuracy (ie lack of bias) and precision have been established.	- Assays are completed in a certified laboratory in Perth BVA. Gold assays are determined using fire assay with 40g charge. Where other elements are assayed using either AAS base metal suite or acid digest with ICP-MS finish. The methods used approach total mineral consumption and are typical of industry standard practice. - No geophysical logging of drilling was performed. - Lab standards, blanks and repeats are included as part of the QAQC system. In addition the laboratory has its own internal QAQC comprising standards, blanks and duplicates. Sample preparation checks of pulverizing at the laboratory include tests to check that the standards of 90% passing 75 micron is being achieved. Follow-up re-assaying is performed by the laboratory upon company request following review of assay data. Acceptable bias and precision is noted in results given the nature of the deposit and the level of classification - RC drill samples from previous owners was fire assay with AAS finish. Review of historic records of received assays confirms this.

Criteria	JORC Code explanation	Commentary
Verification of sampling and assaying	- The verification of significant intersections by either independent or alternative company personnel. - The use of twinned holes. - Documentation of primary data, data entry procedures, data verification, data storage (physical and electronic) protocols. - Discuss any adjustment to assay data.	- Significant intersections are noted in logging and checked with assay results by company personnel both on site and in Perth. - There were a number of holes which overlapped with historic drilling and results appear consistent based on preliminary review of the data. - All primary data is logged digitally on tablet or on paper and later entered into the SQL database. Data is visually checked for errors before being sent to an the companies database manager for further validation and uploaded into an offsite database. Hard copies of original drill logs are kept in onsite office. - Visual checks of the data re completed in Surpac mining software - No adjustments have been made to assay data unless in instances where standard tolerances are not met and reassay is ordered .
Location of data points	- Accuracy and quality of surveys used to locate drill holes (collar and down-hole surveys), trenches, mine workings and other locations used in Mineral Resource estimation. - Specification of the grid system used. - Quality and adequacy of topographic control.	- RC/DD drilling is downhole surveyed utilizing surveyed electronic single shot survey tool at collar, 10 metres then 30m thereafter. No Gyro DH surveys were undertaken on this program. - Surface RC and Diamond drilling is marked out using GPS and final pickups using DGPS collar pickups. - The project lies in MGA 94, zone 52.
Data spacing and distribution	- Data spacing for reporting of Exploration Results. - Whether the data spacing and distribution is sufficient to establish the degree of geological and grade continuity appropriate for the Mineral Resource and Ore Reserve estimation procedure(s) and classifications applied. - Whether sample compositing has been applied.	- Surface diamond drilling in this initial phase has been on a nominal 25-50m along strike spacing. - No compositing is applied to diamond drilling or RC sampling. - Core samples are both sampled to geology of between 0.15 and 1.2m intervals. - RC samples are at 1m interval.
Orientation of data in relation to geological structure	- Whether the orientation of sampling achieves unbiased sampling of possible structures and the extent to which this is known, considering the deposit type. - If the relationship between the drilling orientation and the orientation of key mineralised structures is considered to have introduced a sampling bias, this should be assessed and reported if material.	- No bias of sampling is believed to exist through the drilling orientation - Surface drilling is designed perpendicular to the interpreted orientation of the mineralisation.
Sample security	The measures taken to ensure sample security.	- The chain of custody is managed by Pantoro employees and contractors. Samples are stored on site and delivered in sealed boxes and bags to the lab in Perth - Samples are tracked during shipping.
Audits or reviews	The results of any audits or reviews of sampling techniques and data.	No audit or reviews of sampling techniques have been undertaken however the data is managed by an database consultant who has internal checks/protocols in place.

SECTION 2: REPORTING OF EXPLORATION RESULTS – GRANTS CREEK

Criteria	JORC Code explanation	Commentary
Mineral tenement and land tenure status	- Type, reference name/number, location and ownership including agreements or material issues with third parties such as joint ventures, partnerships, overriding royalties, native title interests, historical sites, wilderness or national park and environmental settings. - The security of the tenure held at the time of reporting along with any known impediments to obtaining a licence to operate in the area.	- The Tenement related to this drilling are 100% held by Pantoro subsidiary company Halls Creek Mining Pty Ltd. This is: E80/4952 - The tenement is in good standing and no known impediments exist.
Exploration done by other parties	Acknowledgment and appraisal of exploration by other parties.	- Excluding the historical mining, the first systematic company based exploration in the region prior to 1980 was completed by Australian Mineral Ventures N.L. with regional mapping and selected rock chips from old workings. Southern ventures continued to explore with phases of more comprehensive regional soil sampling and the completion of 26 RC holes for 636 metres at the known workings - In 1991, Dominion Mining Limited ("Dominion") started work on the area as exploration license E80/1343, with a focus on the historical Kimberley Star mine workings. The company completed reconnaissance mapping, aerial photography, satellite imagery interpretation, rock chip/channel sampling and costeaning. Since 2002, Pacrim Energy Limited has held the tenure over the ground and again commenced work with a review of the historical data. From this work the company recommended that soil sampling, ground magnetic survey, geological mapping and rock- chip sampling be completed. - As JV Partner with Pacrim, Metminco undertook drilling in 2008 and completed 20 holes with 14 of them at the perseverance prospect. The remaining 6 tested other regional targets away from the main trend lines. No significant result was reported in the remaining six holes. - From 1994 - 1997 PMA Gold continued to explore the prospects of Perseverance (E80/1343), Star of Kimberley (M80/366) and Wilsons Reef (M80/233). They considered the mineralisation to be closely associated not only with the structural trends but also with basalts and metasediments of the Biscay Formation. PMA found gold in quartz reefs that occurred as discrete, steeply dipping segregations commonly associated with pyrite and base metal sulphides. - From 1994 - 1997 PMA Gold continued to explore the prospects of Perseverance (E80/1343), Star of Kimberley (M80/366) and Wilsons Reef (M Since 2002, Pacrim Energy Limited has held the tenure over the ground and again commenced work with a review of the historical data. From this work the company recommended that soil sampling, ground magnetic survey, geological mapping and rock- chip sampling be completed. - As JV Partner with Pacrim, Metminco undertook drilling in 2008 and completed 20 holes with 14 of them at the perseverance prospect. The remaining 6 tested other regional targets away from the main trend lines. - Limited work was undertaken by Firestrike up until 2014.80/233).

Criteria	JORC Code explanation	Commentary
Geology	Deposit type, geological setting and style of mineralisation.	- The local geology is summarised as gold hosting quartz reefs within deformed and folded metasedimentary and metavolcanic rocks of Proterozoic age. The oldest rocks of the complex were the Ding Dong Downs Volcanics and the Sophie Downs Granite separated from the overlying Halls Creek Group by an unconformity. - The project area also covers part of the Lower Proterozoic Halls Creek Group sediments and sub-volcanics of the Lamboo Complex whilst the Biscay and overlying Olympio Formations comprise the Upper Halls Creek Group. Overlying this Group, the White Water Volcanics Formation is also present to the east of the Halls Creek Fault Zone, a major structural feature that trends northeast across the Grants Creek leases. - The tenement covers an area of extensive carbonate alteration within greywacke sequences, felsic and mafic volcanics and arkosic arenites in the Halls Creek Mobile Zone. These Lower Proterozoic basic schists and metasediments are considered as the preferential hosts for auriferous quartz/ sulphide lode structures. The mineralized structures lie within an east- northeast trending link formation between two splays of the major regional north-east trending Halls Creek fault Zone. Gold mineralisation occurs in association with silver, lead, zinc and minor copper.
Drill hole Information	- A summary of all information material to the understanding of the exploration results including a tabulation of the following information for all Material drill holes: » easting and northing of the drill hole collar » elevation or RL (Reduced Level – elevation above sea level in metres) of the drill hole collar » dip and azimuth of the hole » down hole length and interception depth » hole length. - If the exclusion of this information is justified on the basis that the information is not Material and this exclusion does not detract from the understanding of the report, the Competent Person should clearly explain why this is the case.	No exploration results are reported as part of this release, results relating to the deposits have been previously released.

Criteria	JORC Code explanation	Commentary
Data aggregation methods	- In reporting Exploration Results, weighting averaging techniques, maximum and/or minimum grade truncations (eg cutting of high grades) and cut-off grades are usually Material and should be stated. - Where aggregate intercepts incorporate short lengths of high grade results and longer lengths of low grade results, the procedure used for such aggregation should be stated and some typical examples of such aggregations should be shown in detail. - The assumptions used for any reporting of metal equivalent values should be clearly stated.	No exploration results are reported as part of this release, results relating to the deposit have been previously released.
Relationship between mineralisation widths and intercept lengths	- These relationships are particularly important in the reporting of Exploration Results. - If the geometry of the mineralisation with respect to the drill hole angle is known, its nature should be reported. - If it is not known and only the down hole lengths are reported, there should be a clear statement to this effect (eg 'down hole length, true width not known').	- Surface DD/RC drilling is perpendicular to the interpreted strike of the mineralisation. - Downhole lengths are reported. - Estimated true widths are not currently known due to the early stage of the drilling with orientations yet to be defined.
Diagrams	Appropriate maps and sections (with scales) and tabulations of intercepts should be included for any significant discovery being reported These should include, but not be limited to a plan view of drill hole collar locations and appropriate sectional views.	No exploration results are reported as part of this release, and therefore no diagrams are included.
Balanced reporting	Where comprehensive reporting of all Exploration Results is not practicable, representative reporting of both low and high grades and/or widths should be practiced to avoid misleading reporting of Exploration Results.	No exploration results are reported as part of this release, results relating to the deposit have been previously released.
Other substantive exploration data	Other exploration data, if meaningful and material, should be reported including (but not limited to): geological observations; geophysical survey results; geochemical survey results; bulk samples – size and method of treatment; metallurgical test results; bulk density, groundwater, geotechnical and rock characteristics; potential deleterious or contaminating substances.	No other meaningful data to report.
Further work	- The nature and scale of planned further work (eg tests for lateral extensions or depth extensions or large-scale step-out drilling). - Diagrams clearly highlighting the areas of possible extensions, including the main geological interpretations and future drilling areas, provided this information is not commercially sensitive.	The results of this maiden Inferred Mineral resource will guide additional infill drilling to guide further Mineral Resource estimates.

SECTION 3: ESTIMATION AND REPORTING OF MINERAL RESOURCES – GRANTS CREEK

Criteria	JORC Code explanation	Commentary
Database integrity	- Measures taken to ensure that data has not been corrupted by, for example, transcription or keying errors, between its initial collection and its use for Mineral Resource estimation purposes. - Data validation procedures used.	- Recent Data input has been governed by lookup tables and programmed import of assay data from lab into database. The database has been checked against the original assay certificates and survey records for completeness and accuracy. - Data was validated by the geologist after input. Data validation checks were carried out by an external database manager in liaison with Pantoro personnel. - Historic drill collars have been picked up by DGPS and all data loaded for spatial validation and compared to metadata recovered from open file reports from previous operators.
Site visits	- Comment on any site visits undertaken by the Competent Person and the outcome of those visits. - If no site visits have been undertaken indicate why this is the case.	The Competent Person, regularly visits the Grants Creek site, inclusive of Perseverance and Star of Kimberley deposits. The CP has a good appreciation of the mineralisation styles comprising Mineral Resources at the Project with regular inspections of drill rigs and drill core, chips during drill programmes, update of surface mapping and commissioning of structural studies.
Geological interpretation	- Confidence in (or conversely, the uncertainty of) the geological interpretation of the mineral deposit. - Nature of the data used and of any assumptions made. - The effect, if any, of alternative interpretations on Mineral Resource estimation. - The use of geology in guiding and controlling Mineral Resource estimation. - The factors affecting continuity both of grade and geology.	- Confidence in the geological interpretation is generally proportional to the drill density with surface mapping utilised to confirm key mineralisation controls, orientation and confidence for the primary mineralised structures. - Data used for the geological interpretation includes surface mapping and drill logging (Reverse Circulation and Diamond Drilling) data. - Alternative interpretations would be unlikely to vary in orientation, volume and width as the primary mineralisation structures are evident in surface outcrop and correlate well in downhole drill data. Thus are currently considered to maintain clear strike and dip continuity and a discrete mineralisation, host rock boundary. The orientation of high grade plunges within the deposits is not currently well understood at this stage of the project and thus an alternative interpretation, which may vary metal distribution locally would not be considered material to global gold content of Maiden Mineral Resources and stage of the Project. - Geological interpretation of the data, with quartz veining as a proxy for mineralisation, was used as a basis for domain interpretations. A nominal cut-off above 0.5 g/t gold was utilised, in combination with geology, for domaining mineralisation zones. - Weathering surfaces were interpreted by PNR geologists from drill logging and extended laterally beyond the limits of the Mineral Resource model. - Geology and grade continuity are constrained by quartz veining within the primary shear zone and parallel structures. At this stage of the project there appears a strong correlation between gold tenor and density of quartz veining.

Criteria	JORC Code explanation	Commentary
Dimensions	The extent and variability of the Mineral Resource expressed as length (along strike or otherwise), plan width, and depth below surface to the upper and lower limits of the Mineral Resource.	- The Perseverance deposit is approximately 260m in strike length and generally 0.5 to 3m wide extending nominally 110 metres below surface. - The Star of Kimberley deposit is approximately 160m in strike length and generally 0.5 to 2m wide extending nominally 70 metres below surface. - Mineralisation within the model which did not satisfy the classification criteria for the MRE remained unclassified.
Estimation and modelling techniques	- The nature and appropriateness of the estimation technique(s) applied and key assumptions, including treatment of extreme grade values, domaining, interpolation parameters and maximum distance of extrapolation from data points. If a computer assisted estimation method was chosen include a description of computer software and parameters used. - The availability of check estimates, previous estimates and/or mine production records and whether the Mineral Resource estimate takes appropriate account of such data. - The assumptions made regarding recovery of by-products. - Estimation of deleterious elements or other non-grade variables of economic significance (e.g. sulphur for acid mine drainage characterisation). - In the case of block model interpolation, the block size in relation to the average sample spacing and the search employed. - Any assumptions behind modelling of selective mining units. - Any assumptions about correlation between variables. - Description of how the geological interpretation was used to control the resource estimates. - Discussion of basis for using or not using grade cutting or capping. - The process of validation, the checking process used, the comparison of model data to drill hole data, and use of reconciliation data if available.	- Separate block models were created for the Perseverance and Star of Kimberley deposits. Mineralised structures were domained separately. Models contain grade estimates and attributes for blocks within each domain only. - Geological interpretation forms the basis for the mineralisation domain wireframes; these were oriented along trends of grade continuity and form hard boundaries during estimation. - Perseverance deposit. Three domains representing the primary mineralisation structure (10) and two parallel mineralised structures (11, 12) were created. - Star of Kimberley. One mineralised domain was delineated for Star of Kimberley, representing the primary mineralisation structure at the deposit (40). - A Check Estimate was undertaken using Inverse Distance Squared (constrained by individual mineralisation domains). - There were no assumptions made with respect to by-products. - No estimation was made for deleterious elements or other non-grade variables. Preliminary gravity and cyanidation metallurgical test work suggests there are not any elements which adversely affect metallurgical recovery. - A 3D volume block model "3DBM" was utilised with all optimised and validated interpolation, density, domains, depletions, classification, and other information required for resource reporting and subsequent mine planning being interpolated and/or available for coding. - Block dimensions for interpolation were Y: 10 mN, X: 10 mE, Z: 10mRL with sub celling of Y: 0.3125 mN, X: 0.3125 mE, Z: 0.3125 mRL to provide adequate domain volume definition and honour wireframe geometry. Considerations relating to appropriate block size include: drill hole data spacing, conceptual mining method, variogram continuity ranges and search neighbourhood optimisation. - Diamond and reverse circulation data was utilised during the estimate. Average sample spacing was 25 metres, which was considered suitable for assessment as Inferred material within a JORC framework. - Minimum and maximum composites for all domains was 4 and 10 respectively. - The Search criteria used for the Perseverance deposit was (Isotropic) 41 m for all three domains, Star of Kimberley deposit was 45 m, with anisotropy ratio of 1.6 in the minor orientation.

Criteria	JORC Code explanation	Commentary
Estimation and modelling techniques		- No selective mining units were assumed in this estimate. Both deposits were considered suitable for potential open pit mining given the grade, depth from surface, mill and comparisons against existing PNR open pit mining at Nicolson's. - No correlated variables have been investigated or estimated. - Apart from the check estimate, all domain estimates were based on mineralisation and geology with domain constraints constructed using a combination of geological logging and a nominal cut-off grade of 0.5 g/t gold. - The mineralisation constraints have been used as hard boundaries for grade estimation wherein only composite samples within that domain are used to estimate blocks coded as within that domain. - Assessment and application of top-cutting for the 3D estimate was undertaken on the gold variable within individual domains. Statistical (and spatial) top cuts were assessed and, where appropriate, were applied on an individual domain basis with one domain at Perseverance capped at 14 g/t gold (domain 10). The remaining two domains at Perseverance remained uncapped. - Statistical and spatial outliers were not identified at Star of Kimberley. No top cuts were applied for the Star of Kimberley mineralisation domain. - Validation of the gold estimate outcomes was completed by global and local bias analysis (swath plots), statistical and visual comparison (cross and long section) with input data.
Moisture	Whether the tonnages are estimated on a dry basis or with natural moisture, and the method of determination of the moisture content	- Tonnage was estimated on a dry basis. - The tonnages of material on stockpiles are quoted on a dry basis.
Cut-off parameters	The basis of the adopted cut-off grade(s) or quality parameters applied	The global gold Inferred Mineral Resource has been reported at a 1.0 g/t gold cut-off and is based upon economic parameters currently utilised at the Nicolson's operations, where deposits of the same style, commodity, comparable size and mining methodology are currently being extracted.
Mining factors or assumptions	Assumptions made regarding possible mining methods, minimum mining dimensions and internal (or, if applicable, external) mining dilution. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider potential mining methods, but the assumptions made regarding mining methods and parameters when estimating Mineral Resources may not always be rigorous. Where this is the case, this should be reported with an explanation of the basis of the mining assumptions made.	The Perseverance and Star of Kimberley MRE's extend nominally 100 m and 65 m, respectively, below topographic surface. Pantoro considers material at this depth would fall within the definition of 'reasonable prospect of eventual economic extraction' within an open pit framework.
Metallurgical factors or assumptions	The basis for assumptions or predictions regarding metallurgical amenability. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider potential metallurgical methods, but the assumptions regarding metallurgical treatment processes and parameters made when reporting Mineral Resources may not always be rigorous. Where this is the case, this should be reported with an explanation of the basis of the metallurgical assumptions made.	Metallurgical test work has shown acceptable (92%) gold recovery using gravity and cyanidation methods which are industry accepted practices. No factors from the metallurgy have been applied to the estimates.

Criteria	JORC Code explanation	Commentary
Environmental factors or assumptions	Assumptions made regarding possible waste and process residue disposal options. It is always necessary as part of the process of determining reasonable prospects for eventual economic extraction to consider the potential environmental impacts of the mining and processing operation. While at this stage the determination of potential environmental impacts, particularly for a greenfields project, may not always be well advanced, the status of early consideration of these potential environmental impacts should be reported. Where these aspects have not been considered this should be reported with an explanation of the environmental assumptions made.	The deposits are located on granted exploration leases with existing historic mining disturbance.
Bulk density	- Whether assumed or determined. If assumed, the basis for the assumptions. If determined, the method used, whether wet or dry, the frequency of the measurements, the nature, size and representativeness of the samples. - The bulk density for bulk material must have been measured by methods that adequately account for void spaces (vugs, porosity, etc.), moisture and differences between rock and alteration zones within the deposit. - Discuss assumptions for bulk density estimates used in the evaluation process of the different materials.	- Bulk density measurements of ore were calculated from drill using the water displacement method. - Bulk densities vary due to ore, weathering type and are assigned separately to each domain based on this work. - Reliable bulk density values are limited within the Grants Creek deposit at this stage of the project and this was taken into account for classification criteria of Mineral Resources. Additional density test work is planned with further drill programmes.
Classification	- The basis for the classification of the Mineral Resources into varying confidence categories. - Whether appropriate account has been taken of all relevant factors (i.e. relative confidence in tonnage/grade estimations, reliability of input data, confidence in continuity of geology and metal values, quality, quantity and distribution of the data). - Whether the result appropriately reflects the Competent Person's view of the deposit.	- This Mineral Resource Estimate has been classified as Inferred to appropriately represent confidence and risk with respect to data quality, drill hole spacing, geological and grade continuity, mineralisation volumes and historical mining activity as well as metal distribution. Additional considerations were the stage of project assessment, amount of diamond drilling, current understanding of mineralisation controls and selectivity within an open pit mining environment. - This approach considers all relevant factors and reflects the Competent Person's view of the deposit.
Audits or reviews	The results of any audits or reviews of Mineral Resource estimates	The current Mineral Resource has been reviewed internally. No reconciliation data exists for this project.
Discussion of relative accuracy/ confidence	- Where appropriate a statement of the relative accuracy and confidence level in the Mineral Resource estimate using an approach or procedure deemed appropriate by the Competent Person. For example, the application of statistical or geostatistical procedures to quantify the relative accuracy of the resource within stated confidence limits, or, if such an approach is not deemed appropriate, a qualitative discussion of the factors that could affect the relative accuracy and confidence of the estimate. - The statement should specify whether it relates to global or local estimates, and, if local, state the relevant tonnages, which should be relevant to technical and economic evaluation. Documentation should include assumptions made and the procedures used. - These statements of relative accuracy and confidence of the estimate should be compared with production data, where available.	- Given the early stage of the project no confidence levels were derived from the current MRE. The relative accuracy of the Mineral Resource estimate is reflected in the reporting of the Mineral Resource as per the guidelines of the 2012 JORC Code. - The statement reflects a global estimate of tonnes and grade. - No production data is available for these deposits.